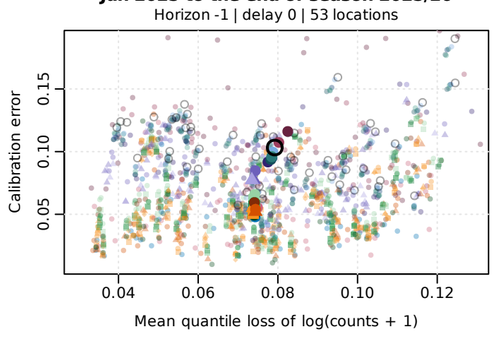
covid | all 53 locations | evaluation: Jun 2025 to the end of season 2025/26 | log1p(counts) updates, loss of log(counts + 1)

Rows: feedback delay (MultiQT learns from the report that many weeks after the first one) and window; columns: horizon. Every delay uses the settings tuned with feedback delay 0.
Small: single locations; large: means over locations; axes: 1st to 99th percentile of both.

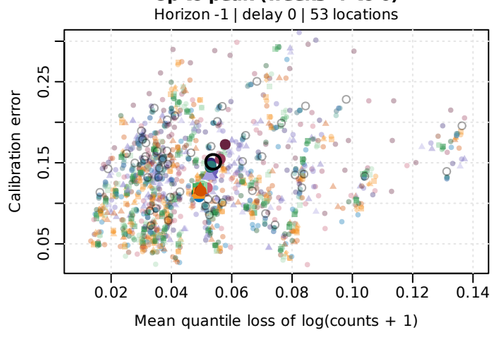
Horizon -1

Feedback delay 0 week(s)

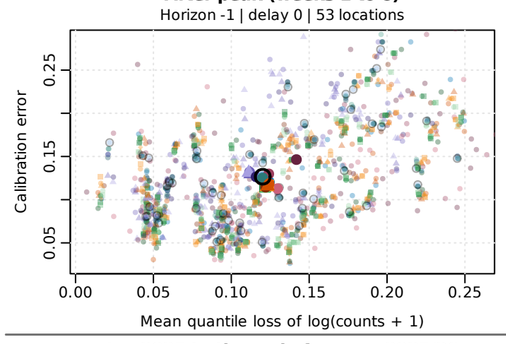
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 0 | 53 locations



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 53 locations

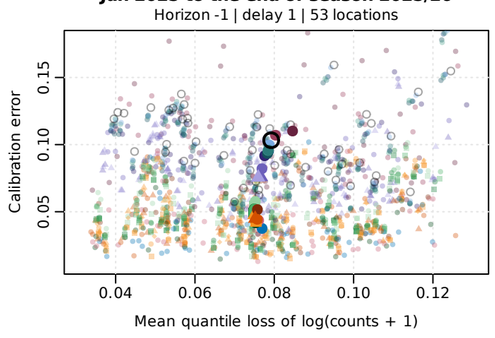


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 53 locations

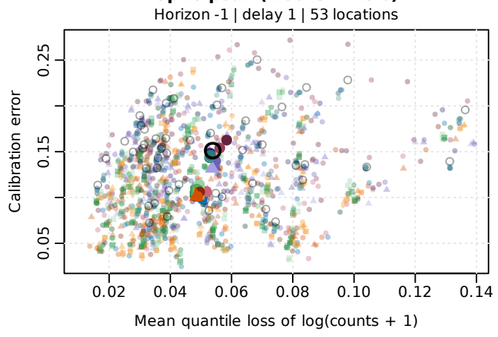


Feedback delay 1 week(s)

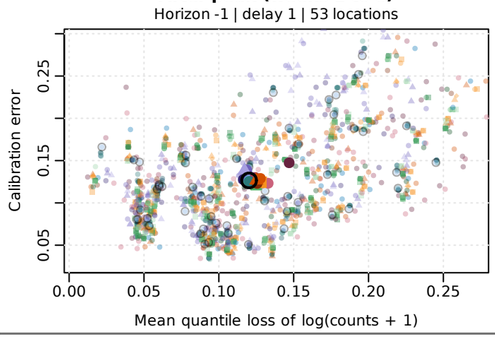
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 1 | 53 locations



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 53 locations

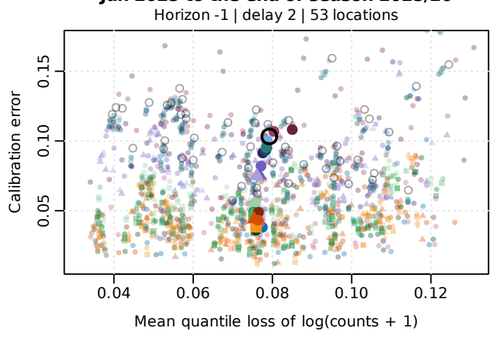


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 53 locations

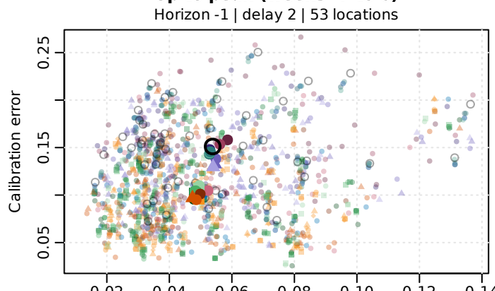


Feedback delay 2 week(s)

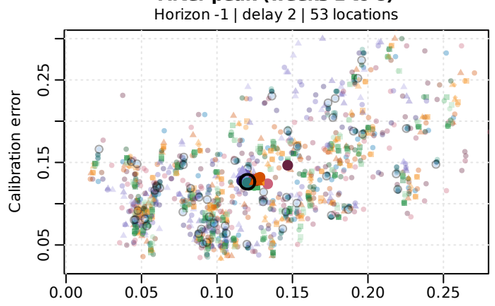
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 2 | 53 locations



Up to peak (weeks -7 to 0)
Horizon -1 | delay 2 | 53 locations

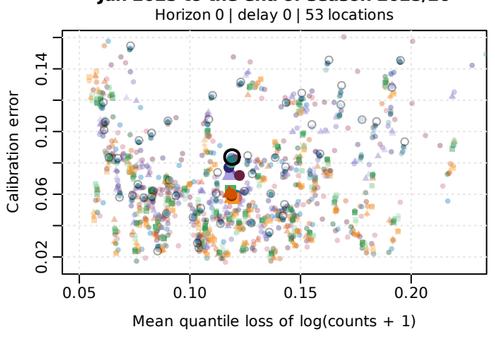


After peak (weeks 1 to 8)
Horizon -1 | delay 2 | 53 locations

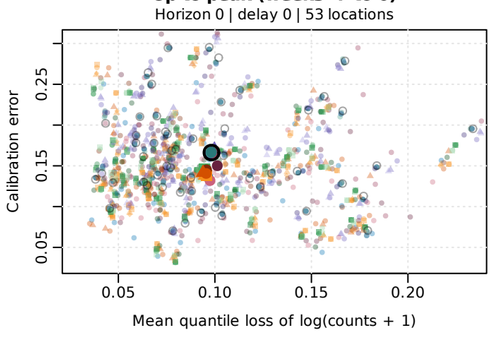


Horizon 0

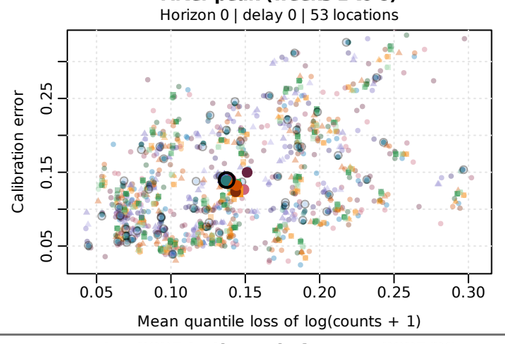
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 0 | 53 locations



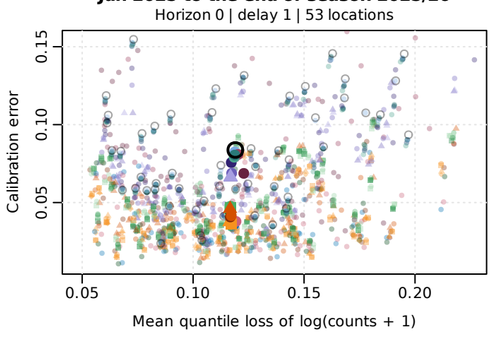
Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 53 locations



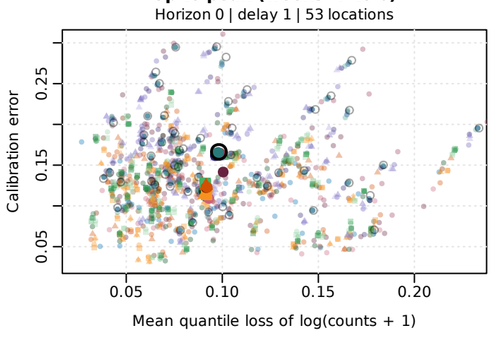
After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 53 locations



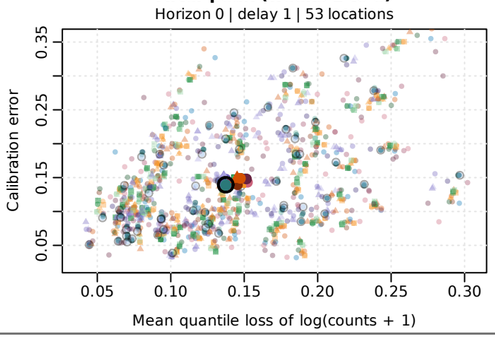
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 1 | 53 locations



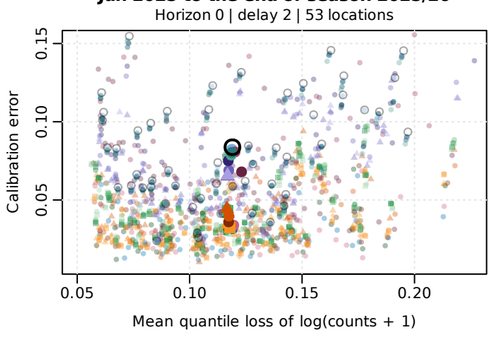
Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 53 locations



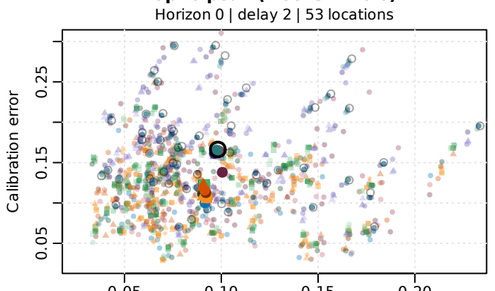
After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 53 locations



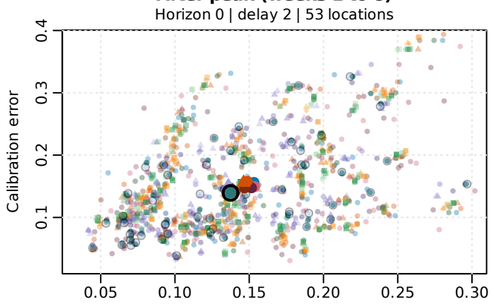
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 2 | 53 locations



Up to peak (weeks -7 to 0)
Horizon 0 | delay 2 | 53 locations

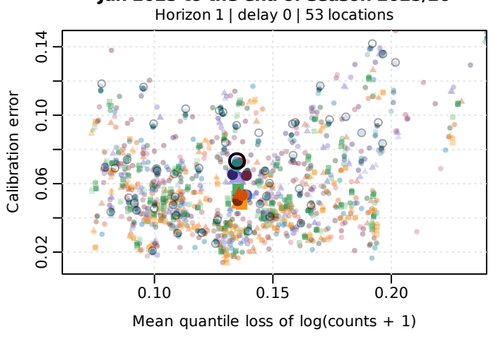


After peak (weeks 1 to 8)
Horizon 0 | delay 2 | 53 locations

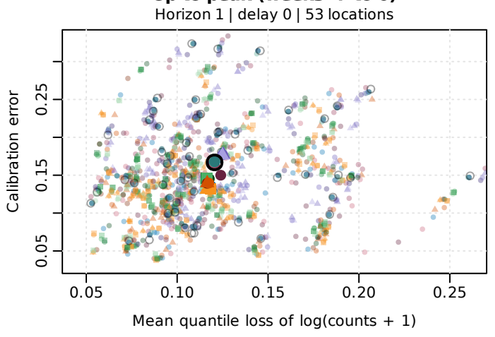


Horizon 1

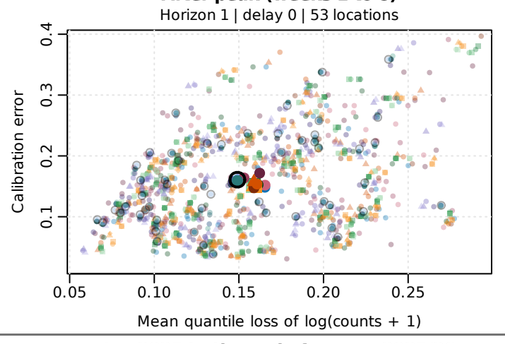
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 0 | 53 locations



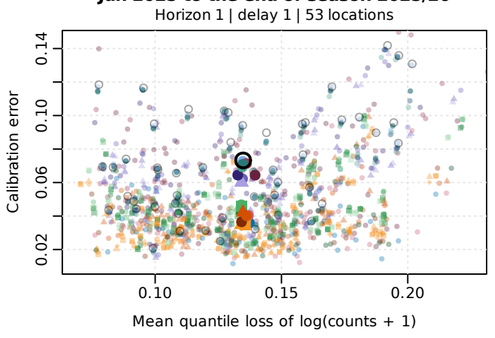
Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 53 locations



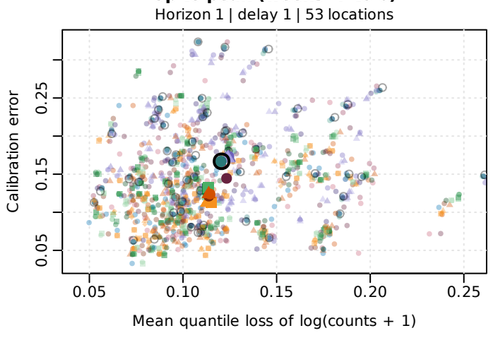
After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 53 locations



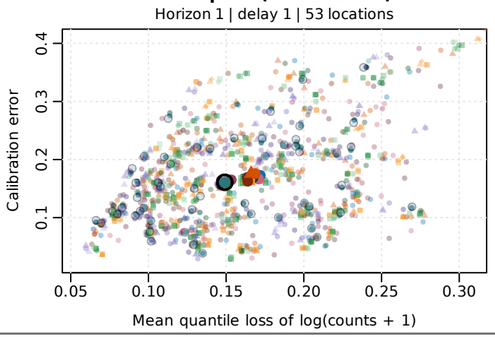
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 1 | 53 locations



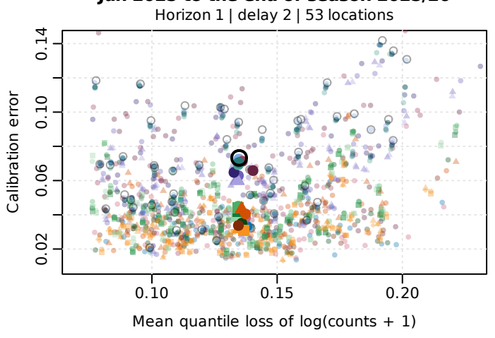
Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 53 locations



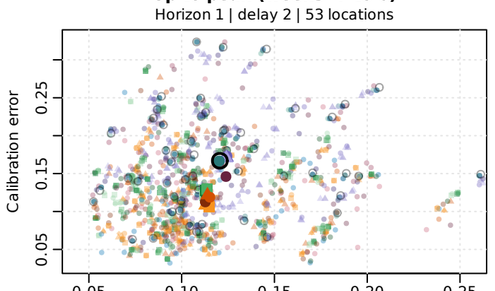
After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 53 locations



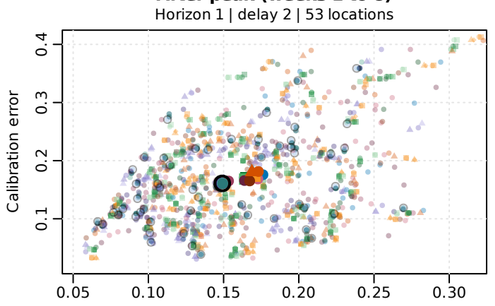
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 2 | 53 locations



Up to peak (weeks -7 to 0)
Horizon 1 | delay 2 | 53 locations

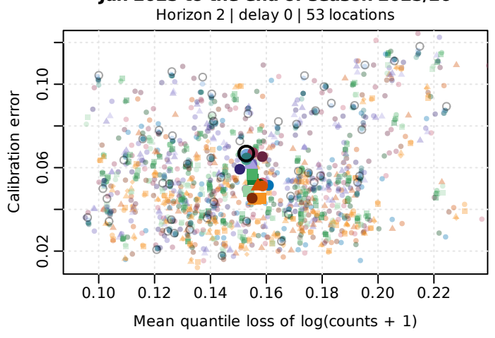


After peak (weeks 1 to 8)
Horizon 1 | delay 2 | 53 locations

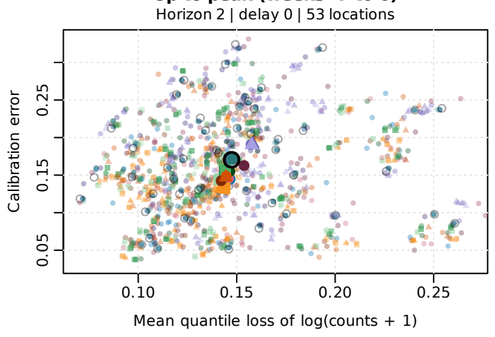


Horizon 2

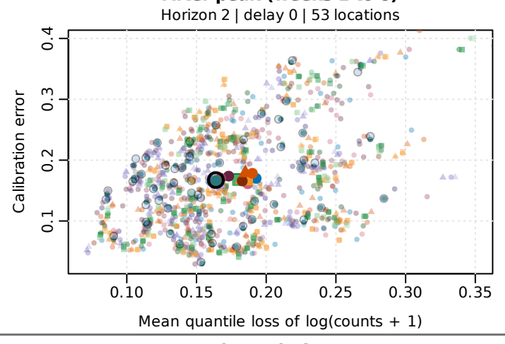
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 0 | 53 locations



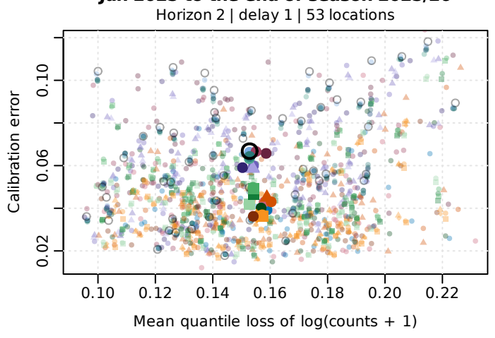
Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 53 locations



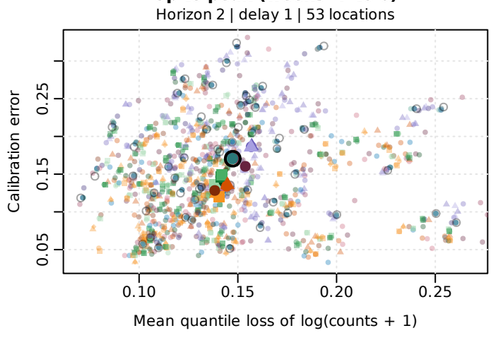
After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 53 locations



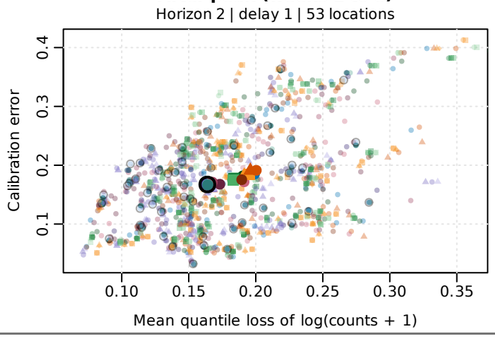
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 1 | 53 locations



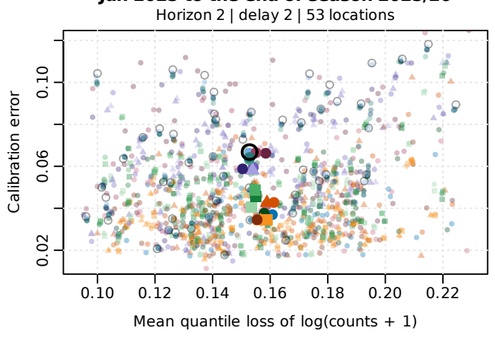
Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 53 locations



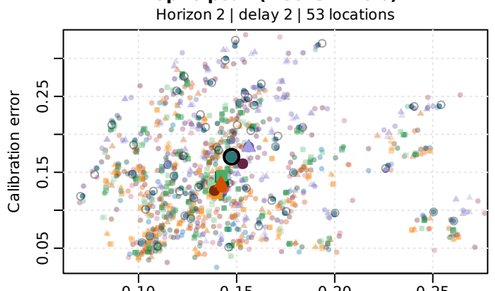
After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 53 locations



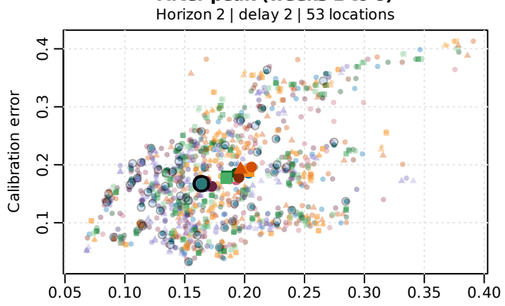
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 2 | 53 locations



Up to peak (weeks -7 to 0)
Horizon 2 | delay 2 | 53 locations

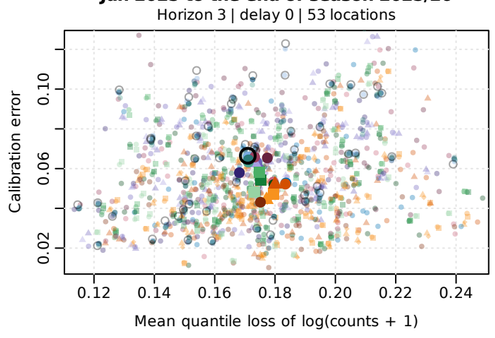


After peak (weeks 1 to 8)
Horizon 2 | delay 2 | 53 locations

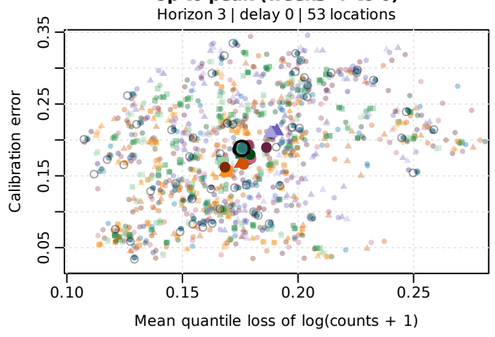


Horizon 3

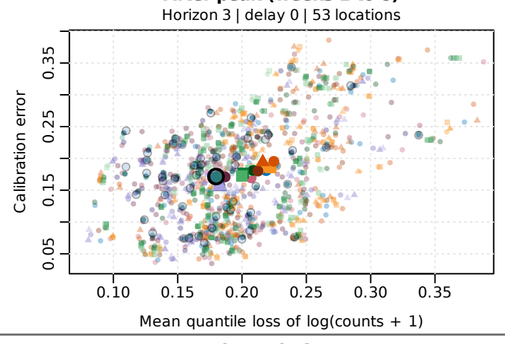
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 0 | 53 locations



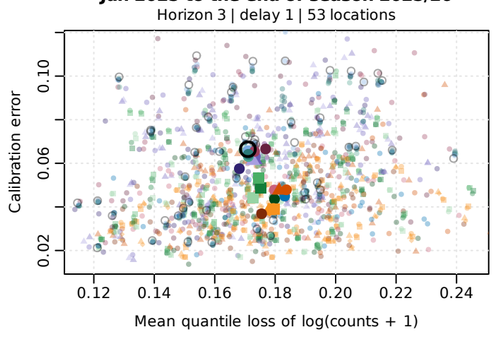
Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 53 locations



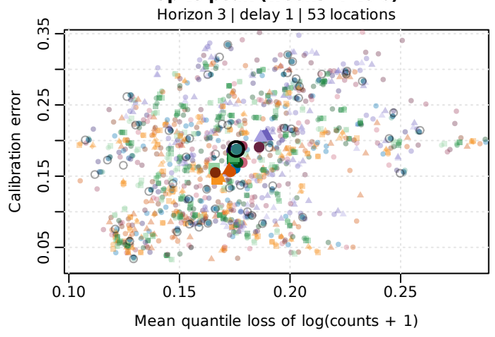
After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 53 locations



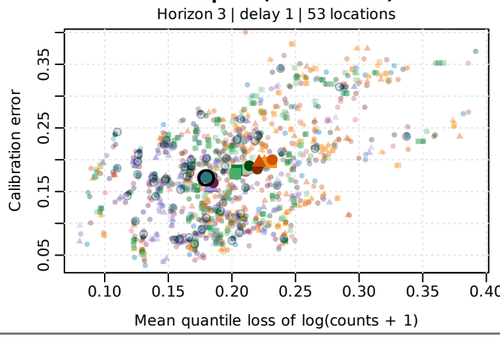
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 1 | 53 locations



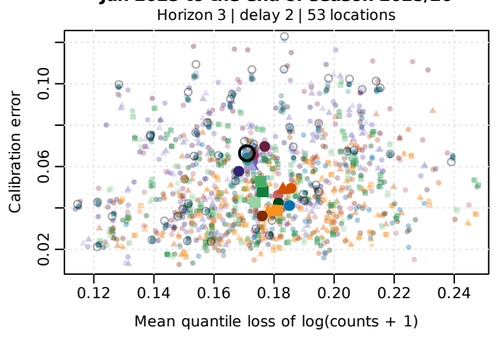
Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 53 locations



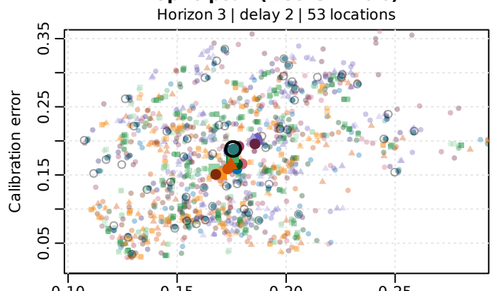
After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 53 locations



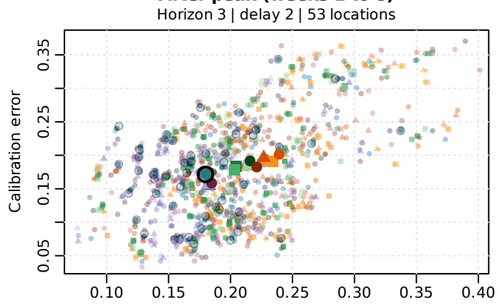
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 2 | 53 locations



Up to peak (weeks -7 to 0)
Horizon 3 | delay 2 | 53 locations



After peak (weeks 1 to 8)
Horizon 3 | delay 2 | 53 locations



Single trackers
- Original
- Fixed MultiQT (eta=0.001*S0)
- Log-slope rate (min m=0.002)
- Target oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=2)
- Rate grid: learned + sharing, common weights (H=2)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Diverse experts
- Diverse experts: equal (slow m=0.001)
- Diverse experts: learned + sharing (slow m=0.001, H=8)
- Diverse experts: learned + sharing, common weights (slow m=0.004)
- Diverse experts: soft switching (strength=1)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.003)
- Studentized median (base 80% width, normal start, eta=0.003)
- Median, constant sigma (RMS of 5, normal start, eta=0.1)

Studentized full base
- Studentized full base (base 80% width, eta=0.003)
- Studentized full base (level scales, eta=0.003)
- Studentized full base (level RMS of 5, eta=0.003)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

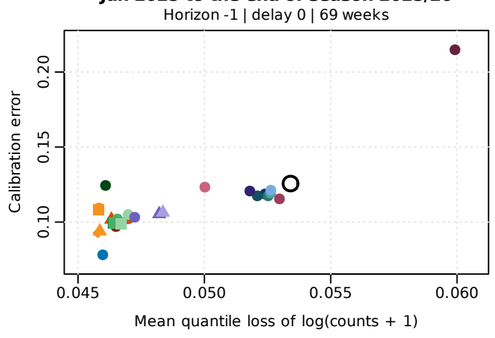
covid | United States (US) | evaluation: Jun 2025 to the end of season 2025/26 | log1p(counts) updates, loss of log(counts + 1)

Rows: feedback delay (MultiQT learns from the report that many weeks after the first one) and window; columns: horizon. Every delay uses the settings tuned with feedback delay 0.

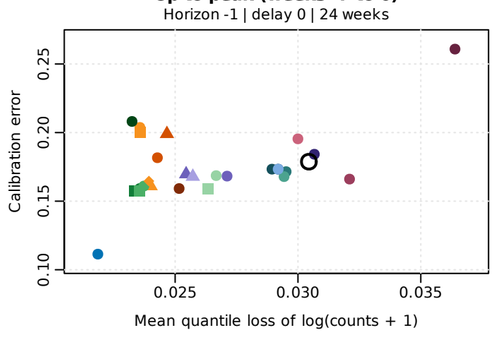
Feedback delay 0 week(s)

Horizon -1

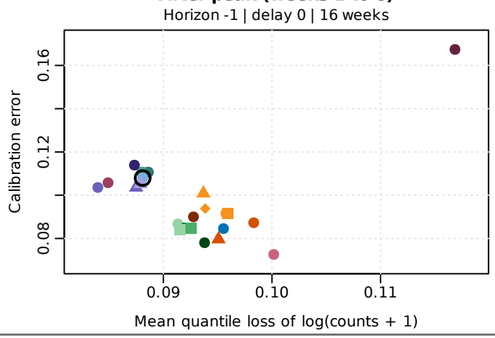
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 0 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 24 weeks

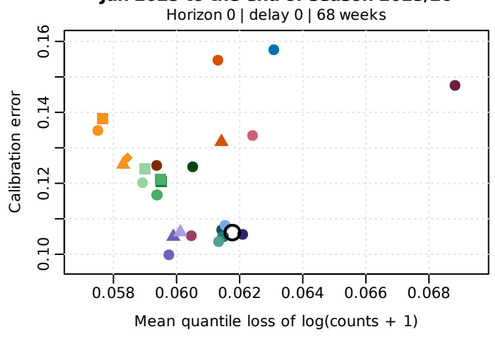


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 16 weeks

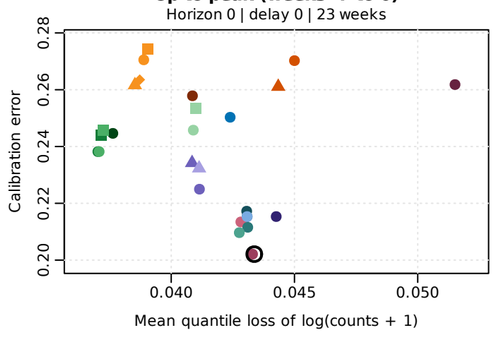


Horizon 0

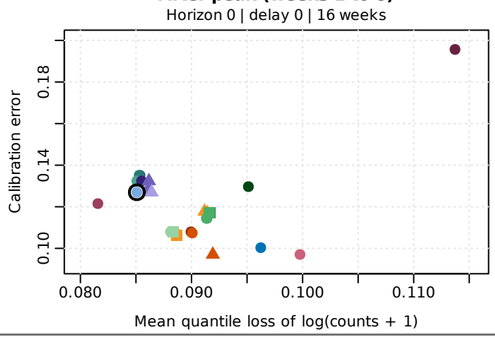
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 0 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 23 weeks

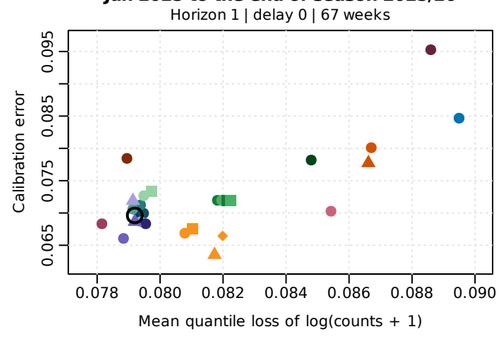


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 16 weeks

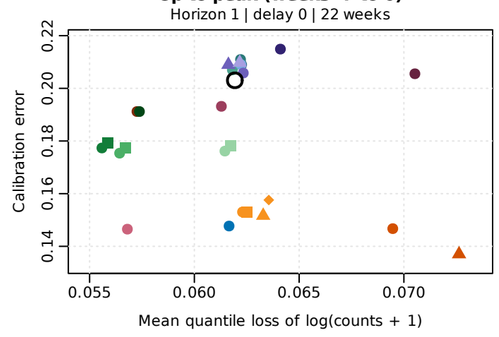


Horizon 1

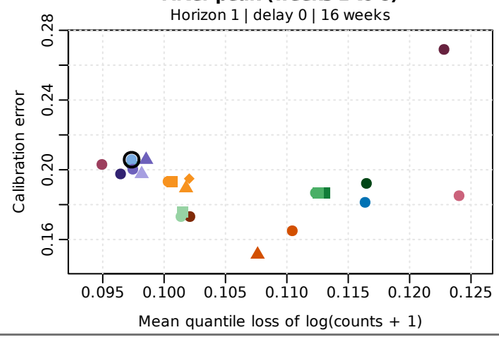
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 0 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 22 weeks

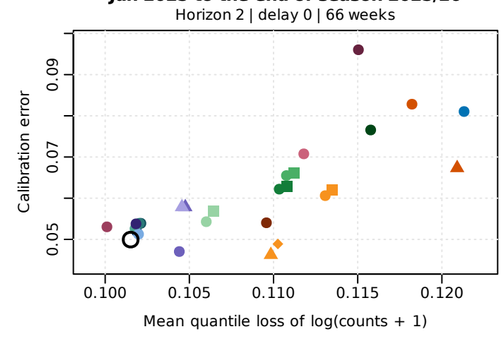


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 16 weeks

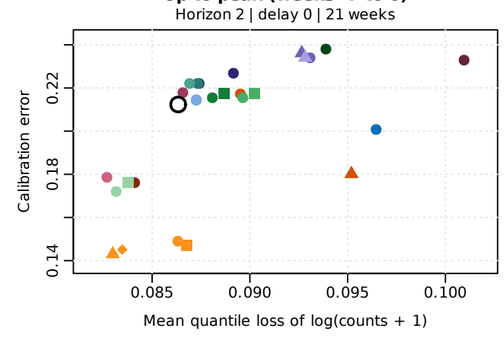


Horizon 2

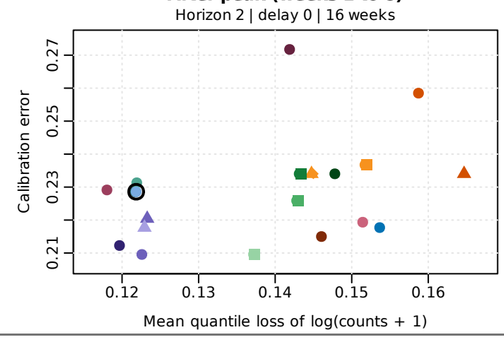
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 0 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 21 weeks

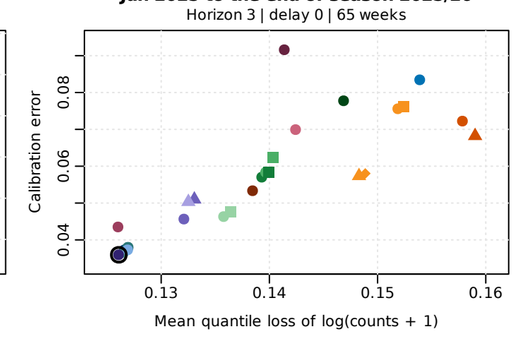


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 16 weeks

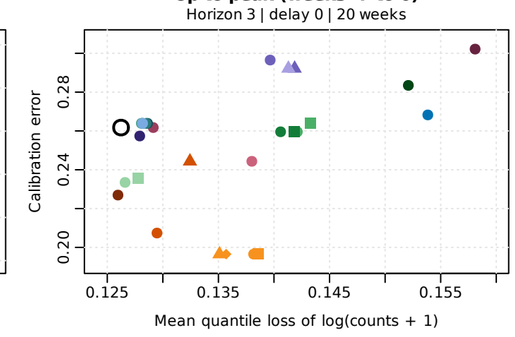


Horizon 3

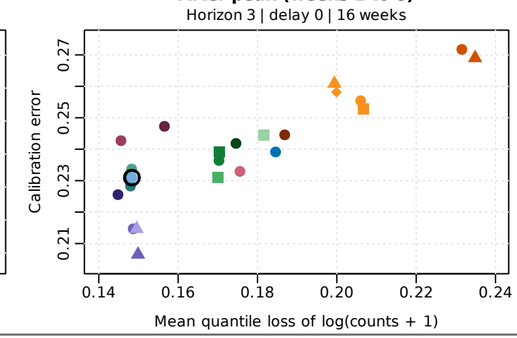
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 0 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 20 weeks



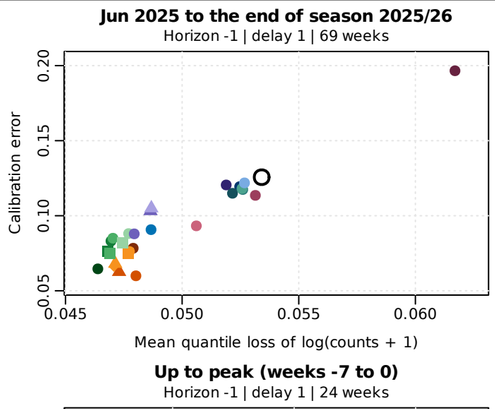
After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 16 weeks



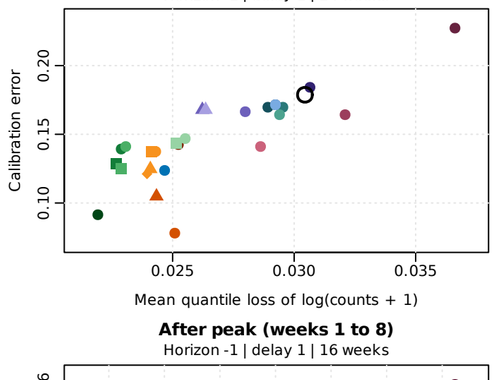
Feedback delay 1 week(s)

Horizon -1

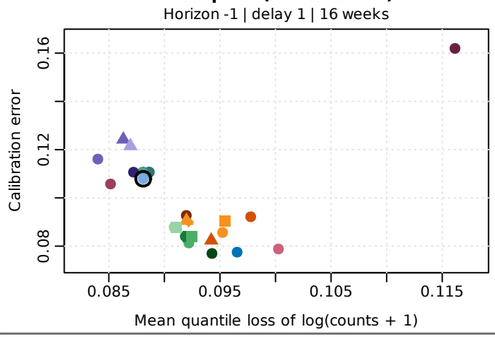
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 1 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 24 weeks

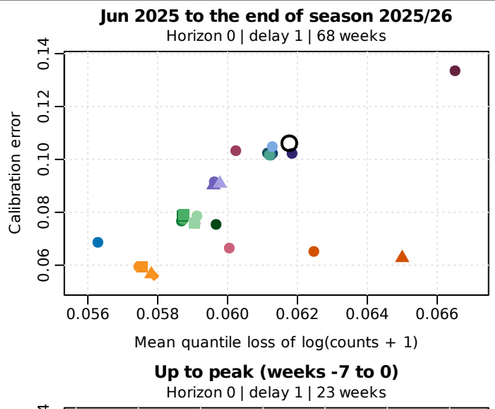


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 16 weeks

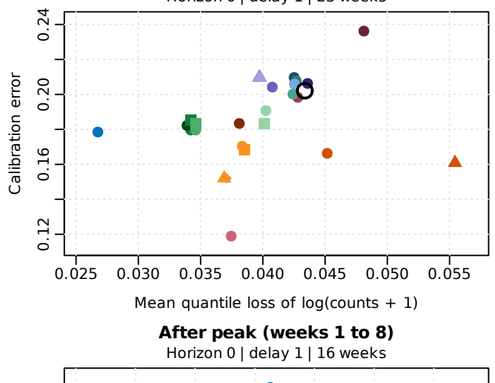


Horizon 0

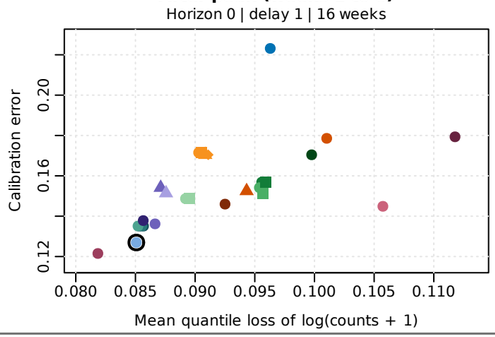
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 1 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 23 weeks

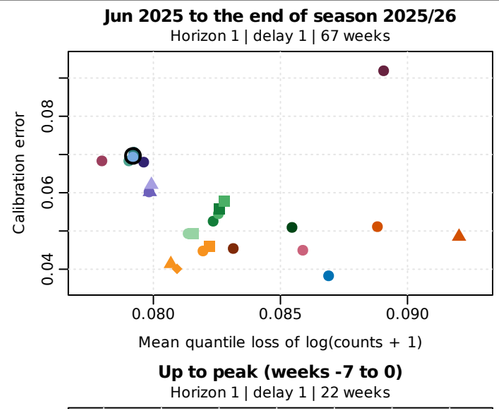


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 16 weeks

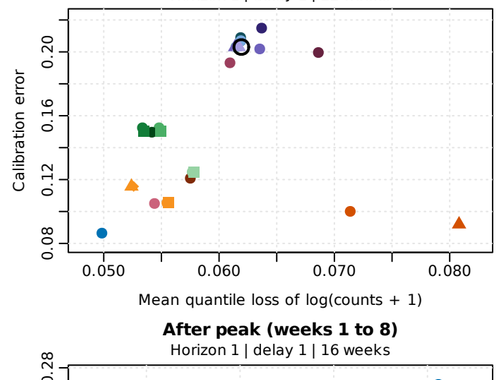


Horizon 1

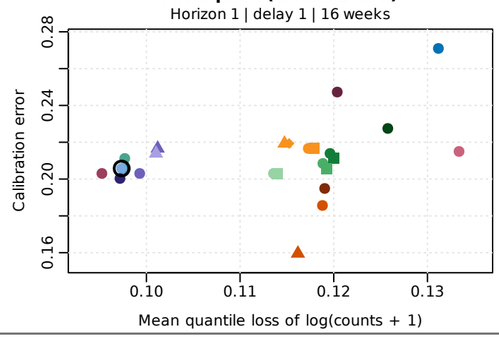
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 1 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 22 weeks

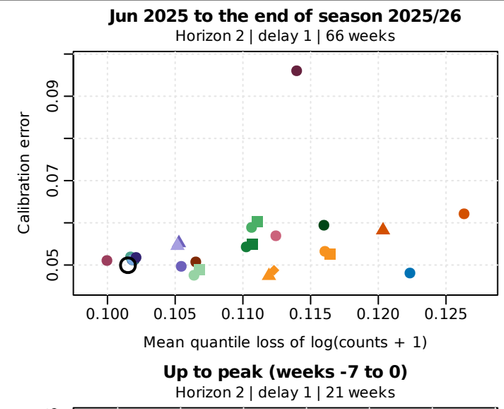


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 16 weeks

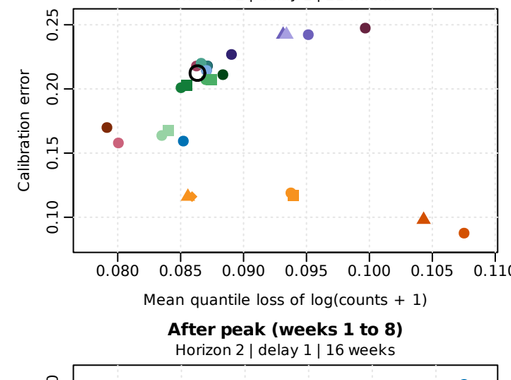


Horizon 2

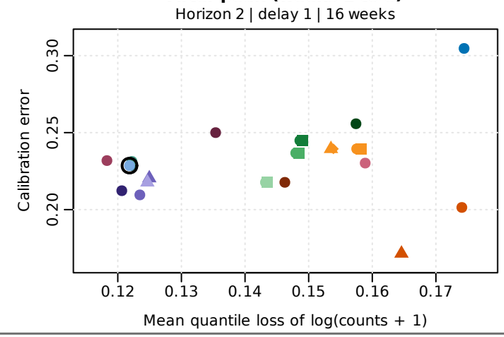
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 1 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 21 weeks

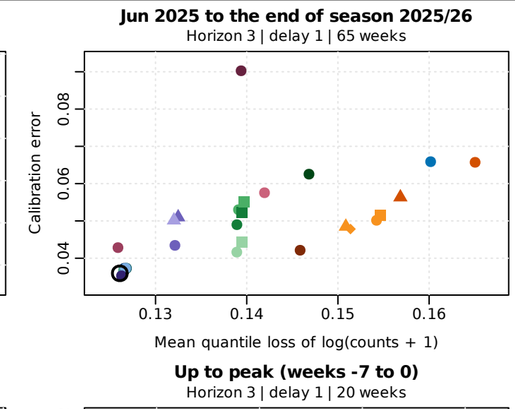


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 16 weeks

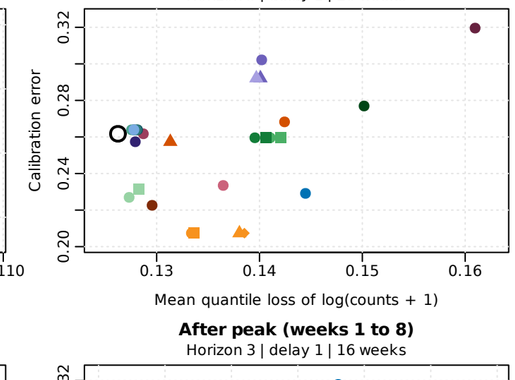


Horizon 3

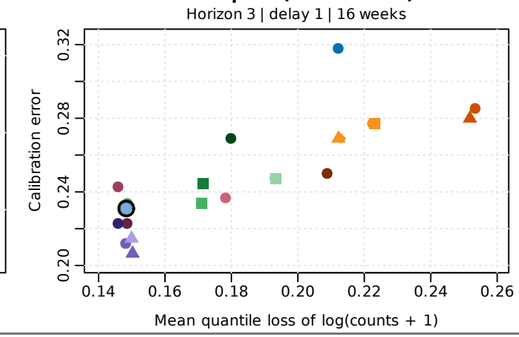
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 1 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 20 weeks



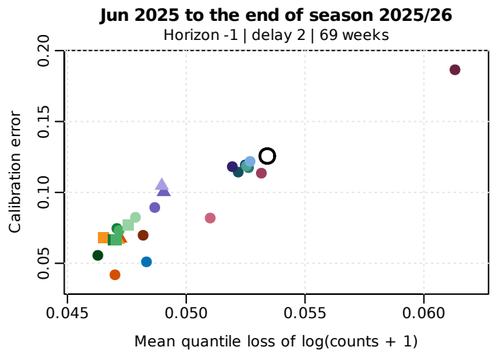
After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 16 weeks



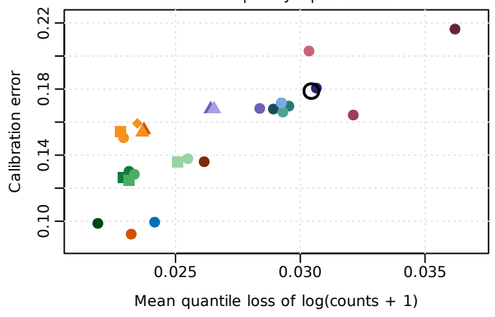
Feedback delay 2 week(s)

Horizon -1

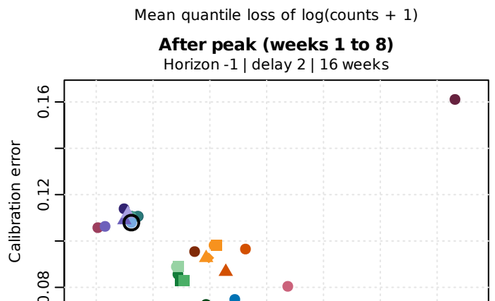
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 2 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 2 | 24 weeks

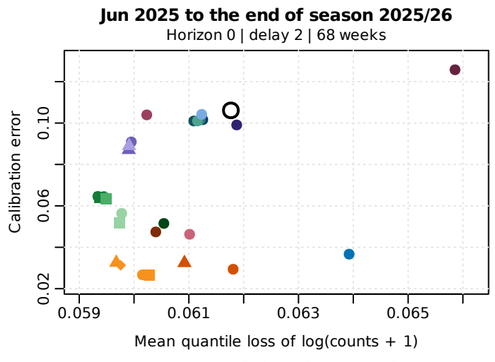


After peak (weeks 1 to 8)
Horizon -1 | delay 2 | 16 weeks

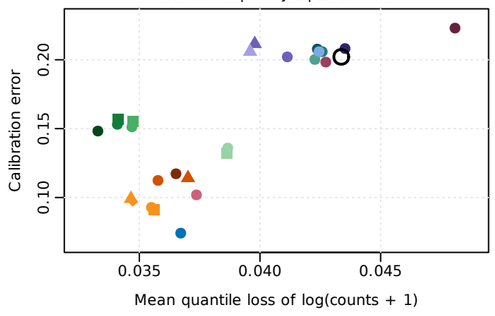


Horizon 0

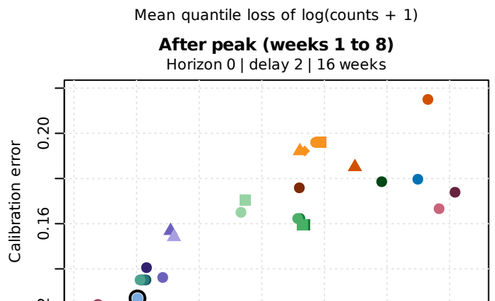
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 2 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 2 | 23 weeks

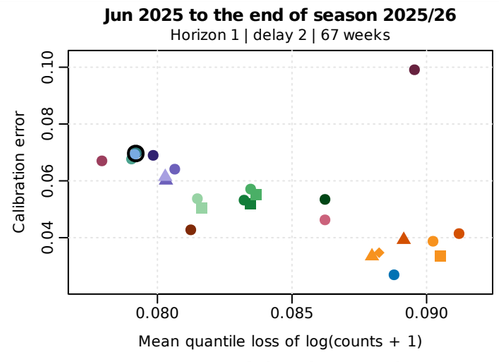


After peak (weeks 1 to 8)
Horizon 0 | delay 2 | 16 weeks

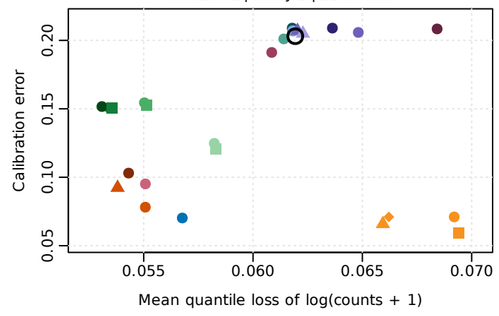


Horizon 1

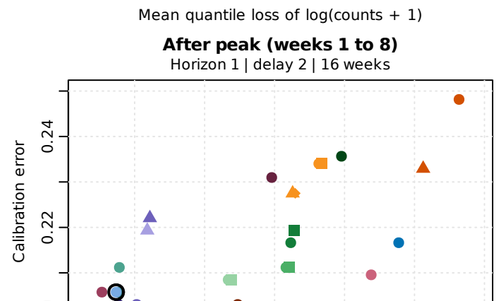
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 2 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 2 | 22 weeks

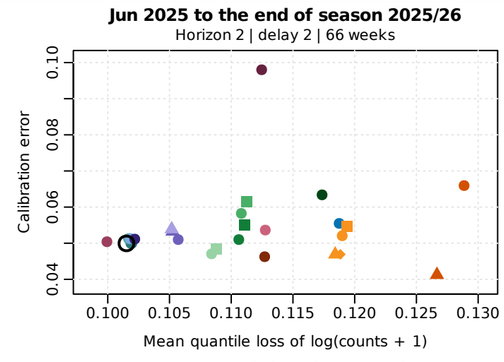


After peak (weeks 1 to 8)
Horizon 1 | delay 2 | 16 weeks

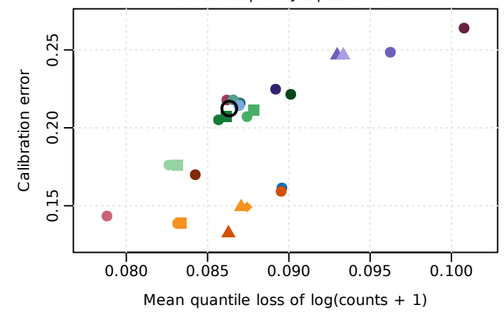


Horizon 2

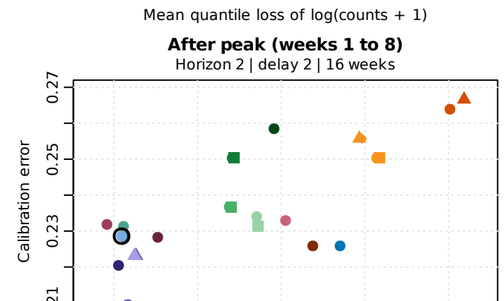
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 2 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 2 | 21 weeks

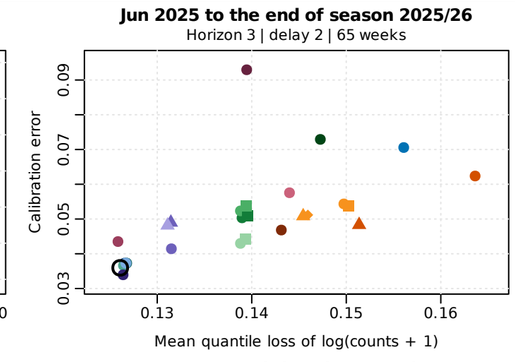


After peak (weeks 1 to 8)
Horizon 2 | delay 2 | 16 weeks

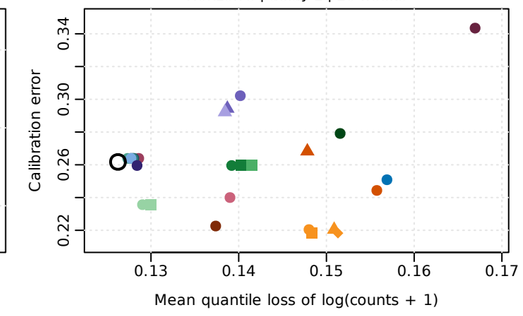


Horizon 3

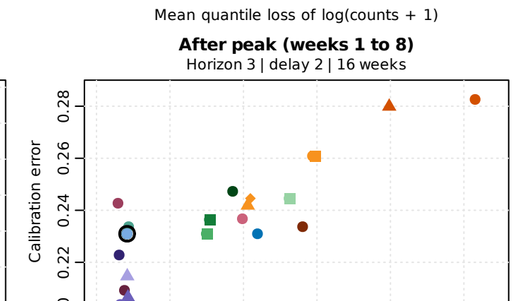
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 2 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 2 | 20 weeks



After peak (weeks 1 to 8)
Horizon 3 | delay 2 | 16 weeks



Single trackers
- Original
- Fixed MultiQT (eta=0.001*S0)
- Log-slope rate (min m=0.002)
- Target oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=2)
- Rate grid: learned + sharing, common weights (H=2)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Diverse experts
- Diverse experts: equal (slow m=0.001)
- Diverse experts: learned + sharing (slow m=0.001, H=8)
- Diverse experts: learned + sharing, common weights (slow m=0.004)
- Diverse experts: soft switching (strength=1)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.003)
- Studentized median (base 80% width, normal start, eta=0.003)
- Median, constant sigma (RMS of 5, normal start, eta=0.1)

Studentized full base
- Studentized full base (base 80% width, eta=0.003)
- Studentized full base (level scales, eta=0.003)
- Studentized full base (level RMS of 5, eta=0.003)

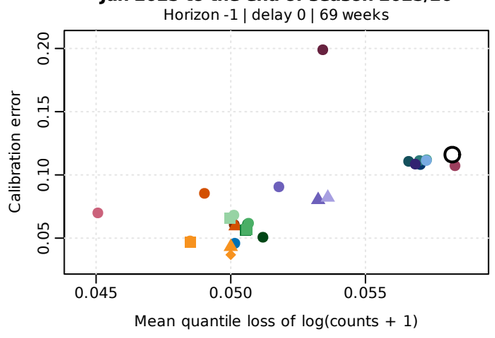
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | California (06) | evaluation: Jun 2025 to the end of season 2025/26 | log1p(counts) updates, loss of log(counts + 1)

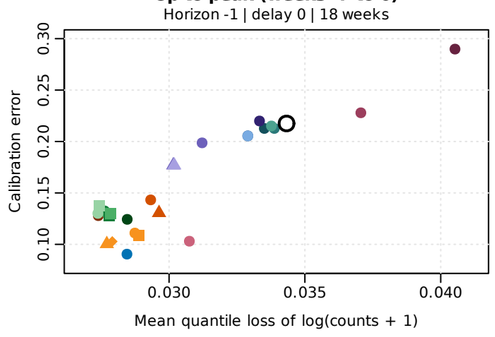
Rows: feedback delay (MultiQT learns from the report that many weeks after the first one) and window; columns: horizon. Every delay uses the settings tuned with feedback delay 0.

Horizon -1

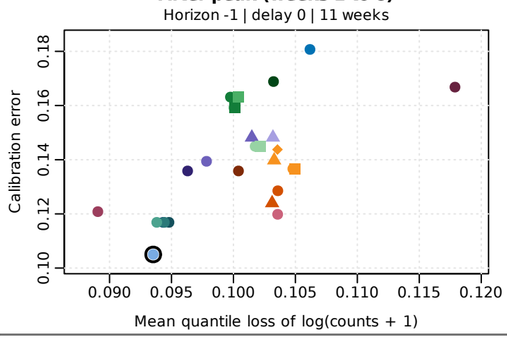
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 0 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 18 weeks

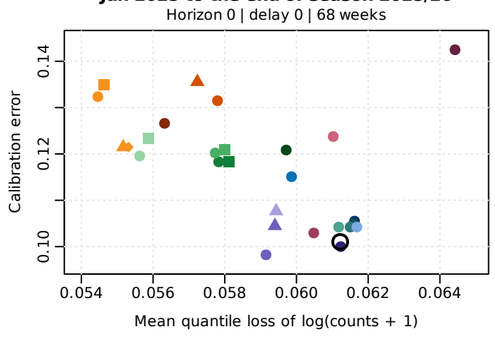


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 11 weeks

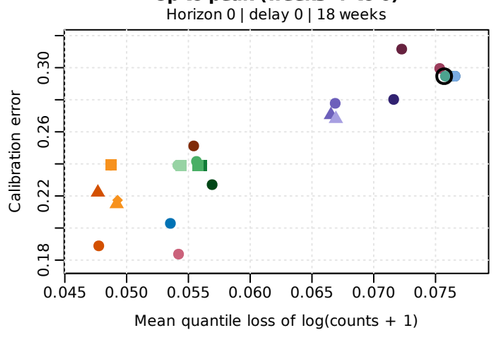


Horizon 0

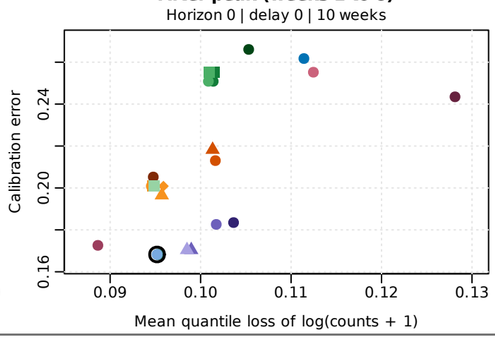
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 0 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 18 weeks

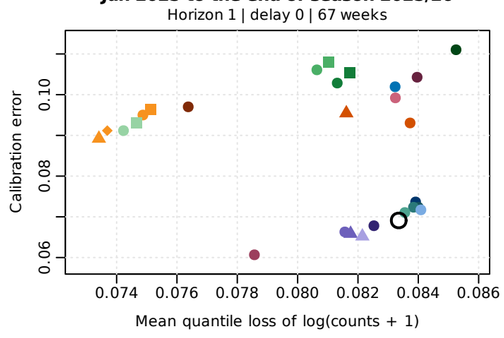


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 10 weeks

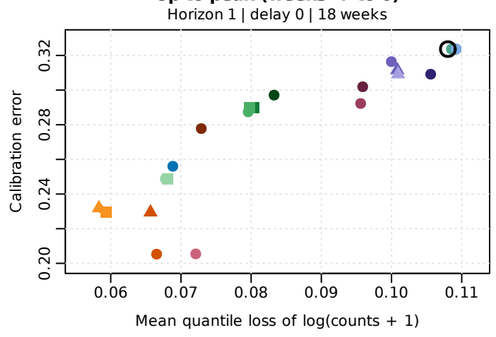


Horizon 1

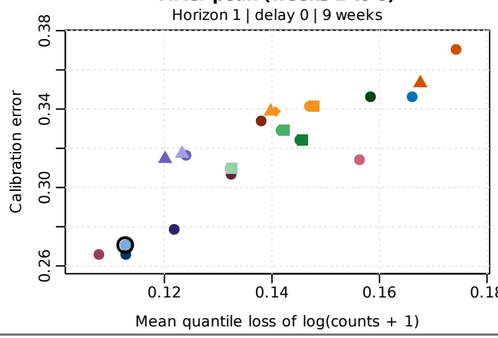
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 0 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 18 weeks

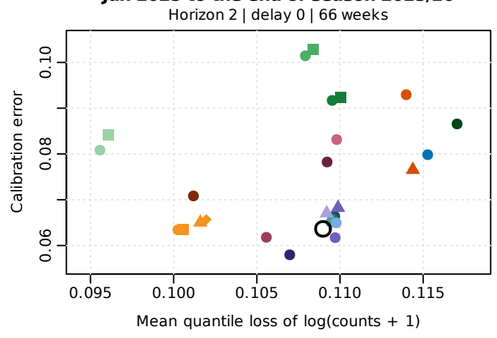


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 9 weeks

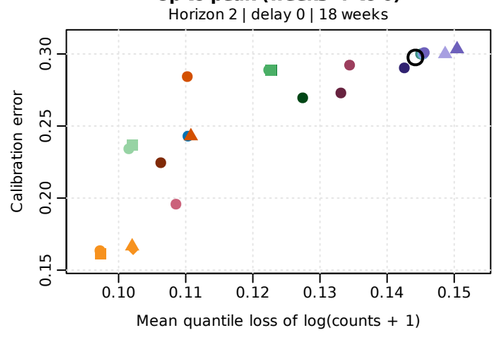


Horizon 2

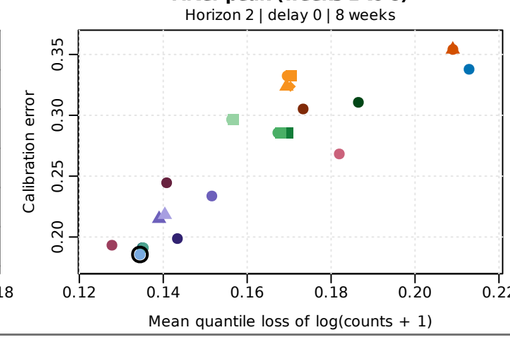
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 0 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 18 weeks

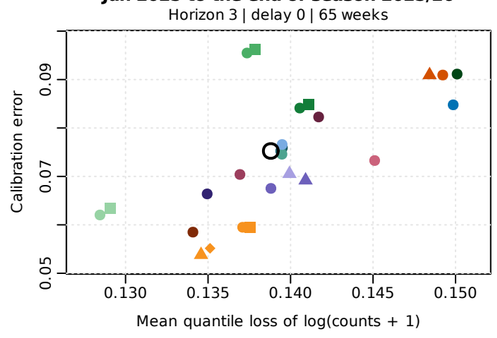


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 8 weeks

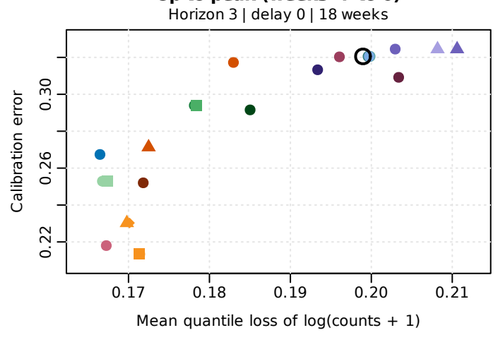


Horizon 3

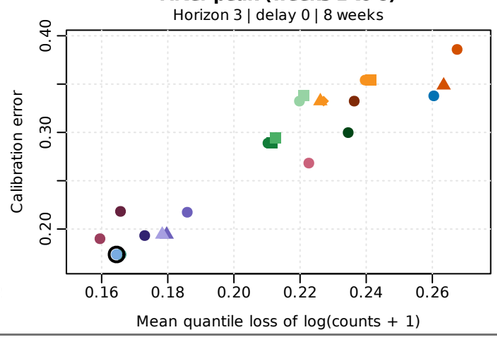
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 0 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 18 weeks



After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 8 weeks

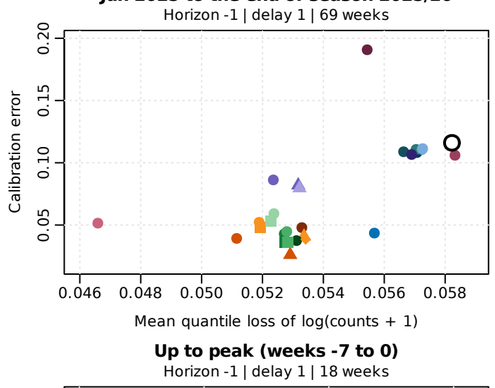


Feedback delay 0 week(s)

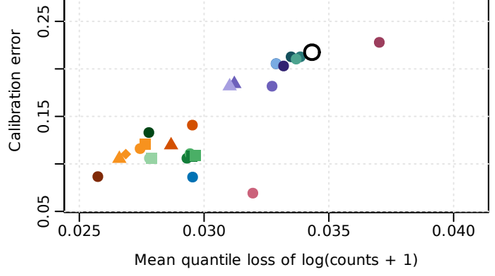
Feedback delay 1 week(s)

Horizon -1

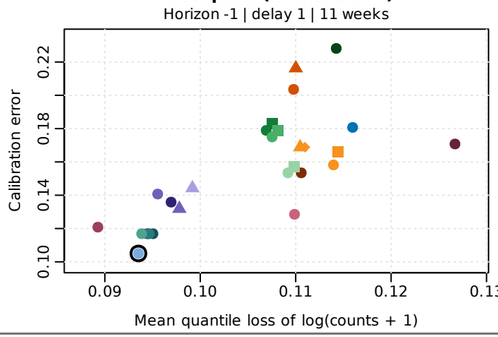
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 1 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 18 weeks

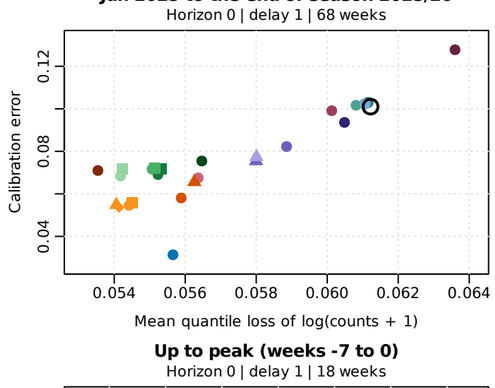


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 11 weeks

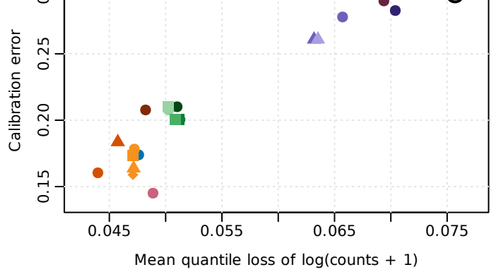


Horizon 0

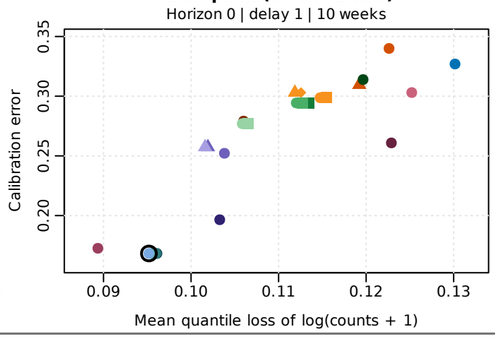
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 1 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 18 weeks

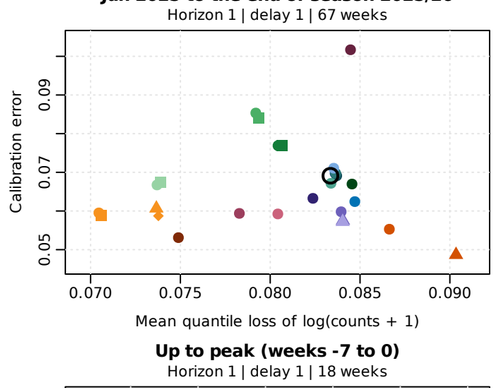


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 10 weeks

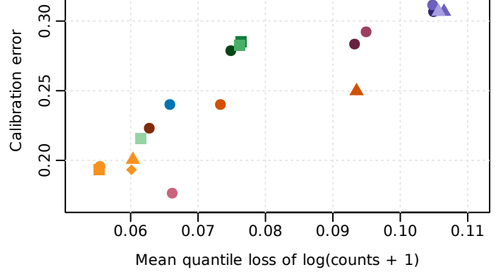


Horizon 1

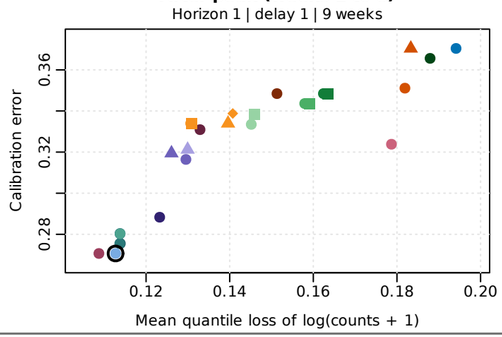
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 1 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 18 weeks

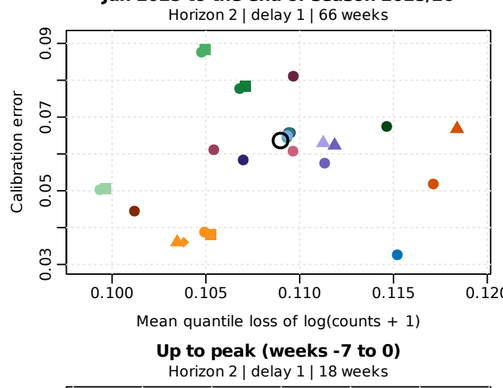


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 9 weeks

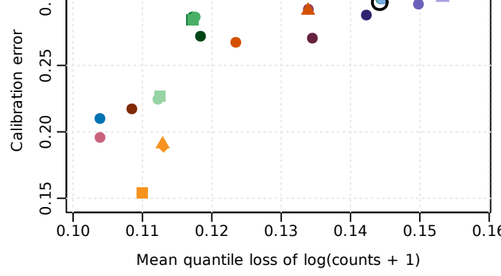


Horizon 2

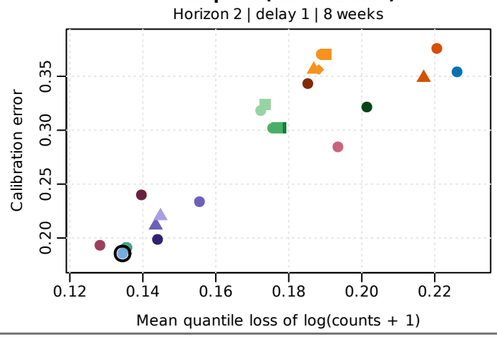
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 1 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 18 weeks

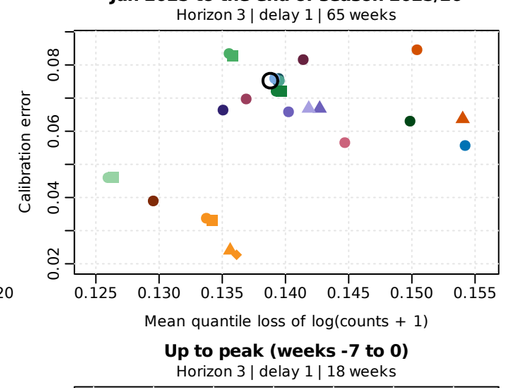


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 8 weeks

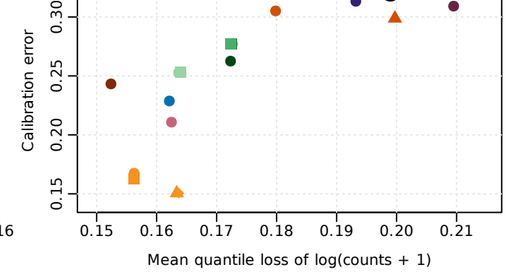


Horizon 3

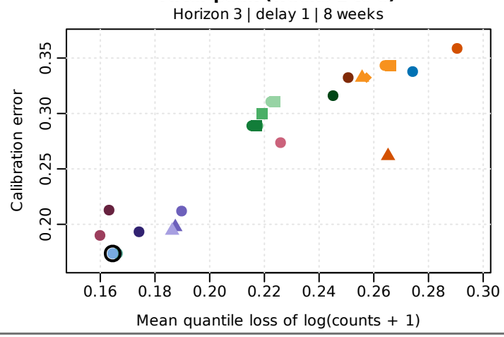
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 1 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 18 weeks



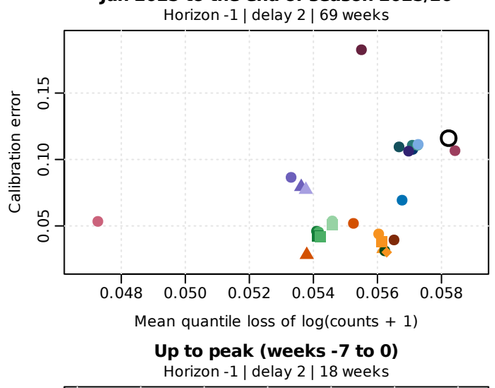
After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 8 weeks



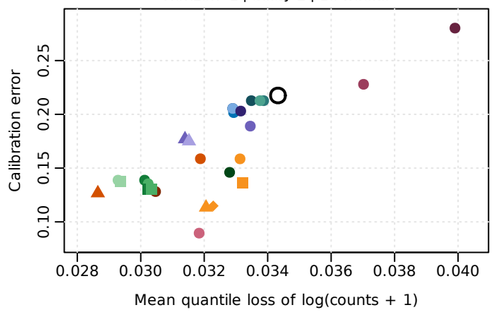
Feedback delay 2 week(s)

Horizon -1

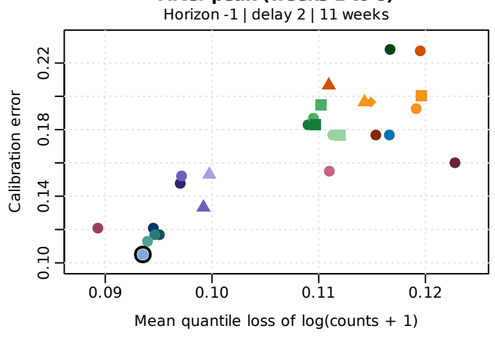
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 2 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 2 | 18 weeks

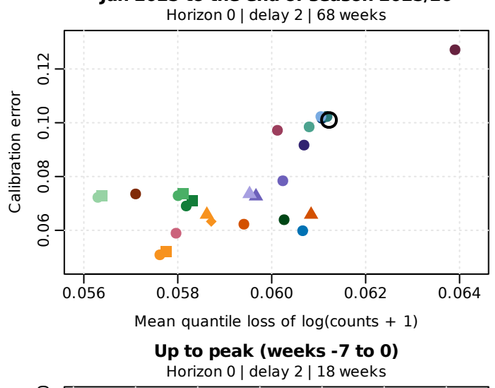


After peak (weeks 1 to 8)
Horizon -1 | delay 2 | 11 weeks

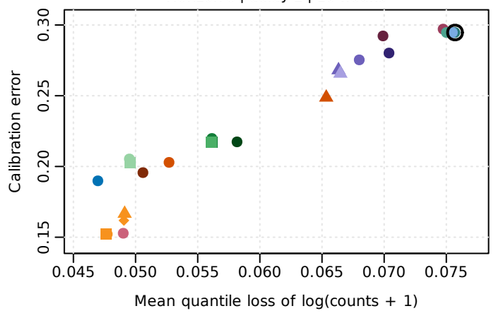


Horizon 0

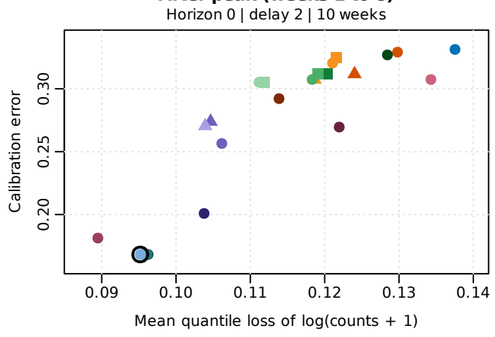
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 2 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 2 | 18 weeks

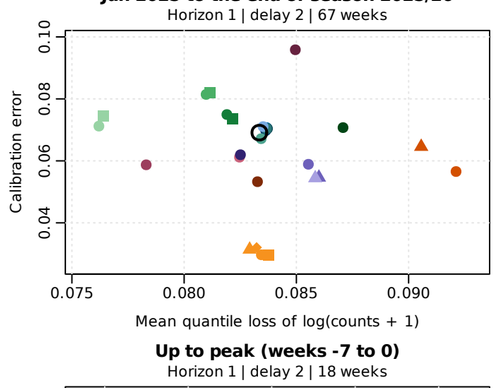


After peak (weeks 1 to 8)
Horizon 0 | delay 2 | 10 weeks

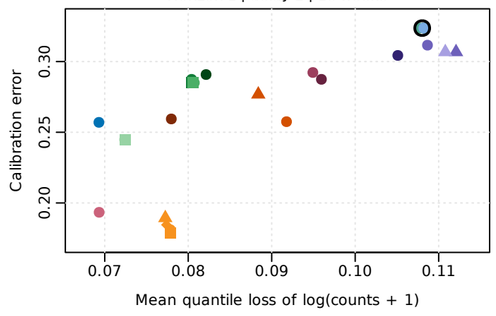


Horizon 1

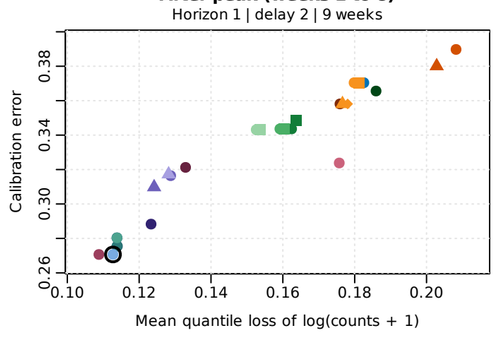
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 2 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 2 | 18 weeks

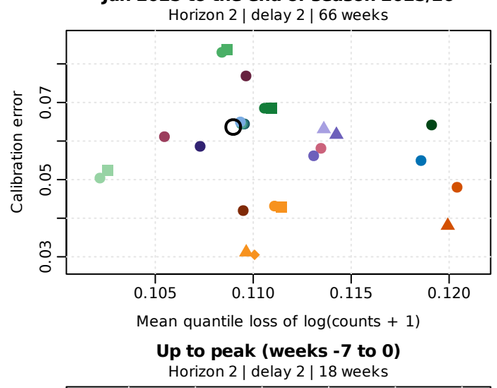


After peak (weeks 1 to 8)
Horizon 1 | delay 2 | 9 weeks

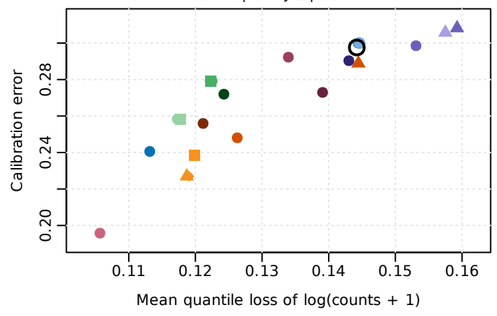


Horizon 2

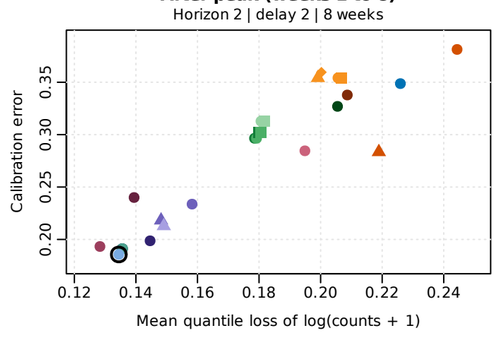
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 2 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 2 | 18 weeks

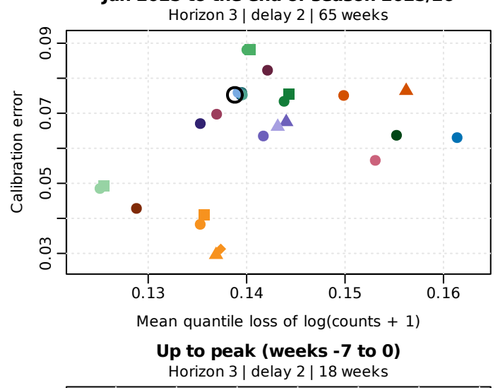


After peak (weeks 1 to 8)
Horizon 2 | delay 2 | 8 weeks

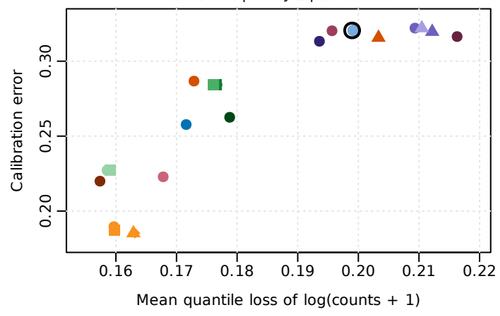


Horizon 3

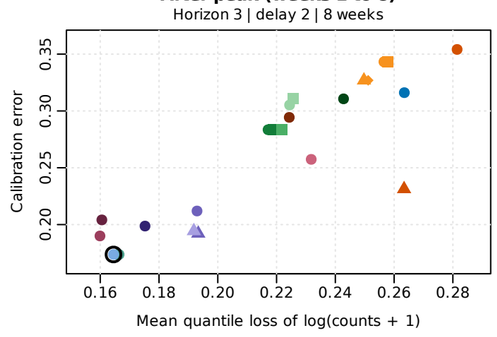
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 2 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 2 | 18 weeks



After peak (weeks 1 to 8)
Horizon 3 | delay 2 | 8 weeks



Single trackers
- Original
- Fixed MultiQT (eta=0.001*S0)
- Log-slope rate (min m=0.002)
- Target oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=2)
- Rate grid: learned + sharing, common weights (H=2)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Diverse experts
- Diverse experts: equal (slow m=0.001)
- Diverse experts: learned + sharing (slow m=0.001, H=8)
- Diverse experts: learned + sharing, common weights (slow m=0.004)
- Diverse experts: soft switching (strength=1)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.003)
- Studentized median (base 80% width, normal start, eta=0.003)
- Median, constant sigma (RMS of 5, normal start, eta=0.1)

Studentized full base
- Studentized full base (base 80% width, eta=0.003)
- Studentized full base (level scales, eta=0.003)
- Studentized full base (level RMS of 5, eta=0.003)

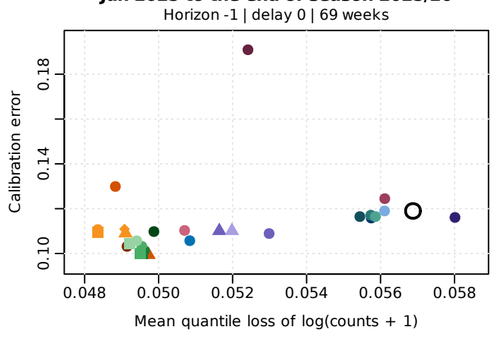
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | Texas (48) | evaluation: Jun 2025 to the end of season 2025/26 | log1p(counts) updates, loss of log(counts + 1)

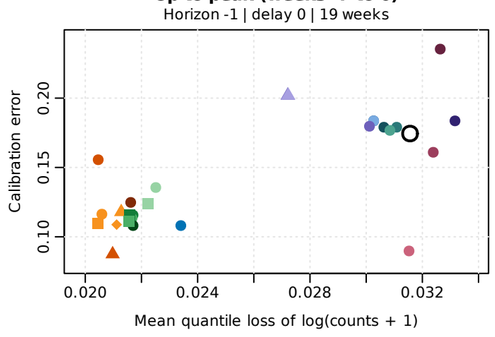
Rows: feedback delay (MultiQT learns from the report that many weeks after the first one) and window; columns: horizon. Every delay uses the settings tuned with feedback delay 0.

Horizon -1

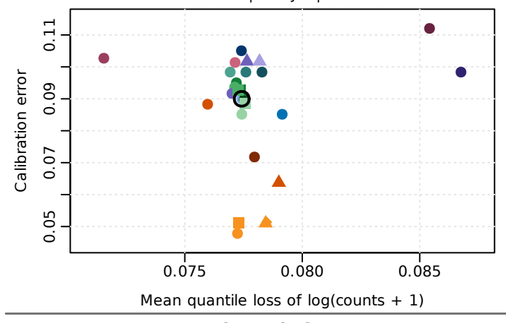
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 0 | 69 weeks



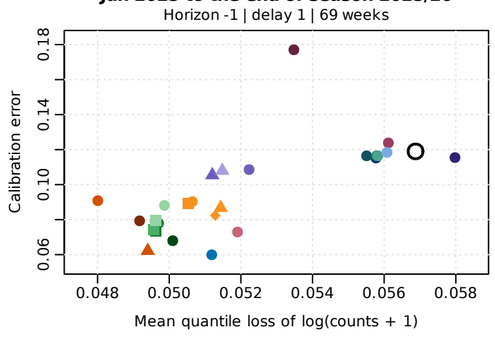
Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 19 weeks



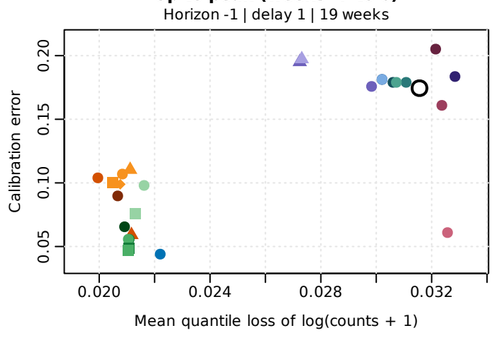
After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 13 weeks



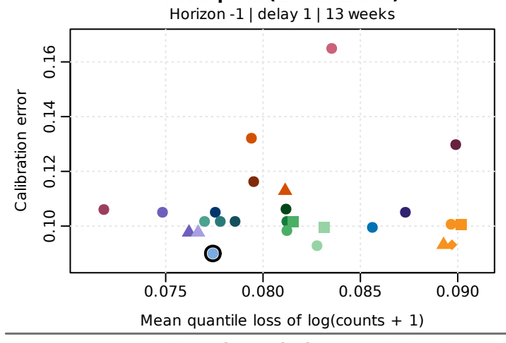
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 1 | 69 weeks



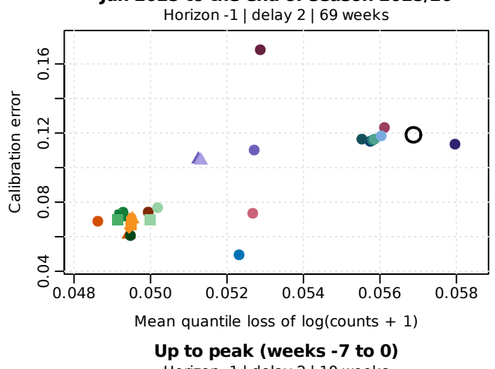
Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 19 weeks



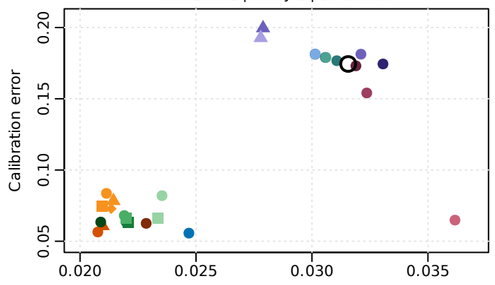
After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 13 weeks



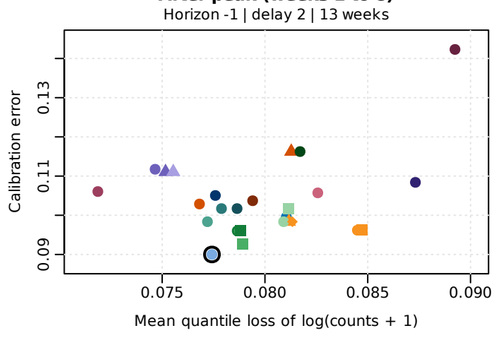
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 2 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 2 | 19 weeks

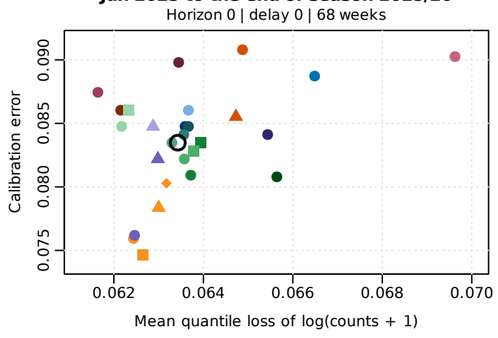


After peak (weeks 1 to 8)
Horizon -1 | delay 2 | 13 weeks

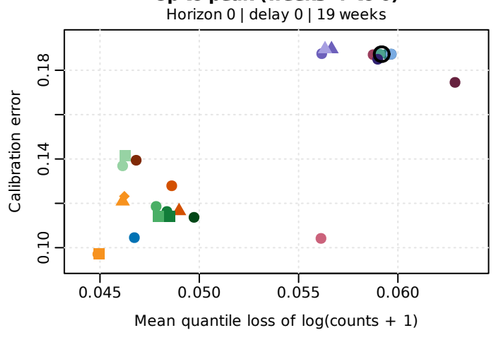


Horizon 0

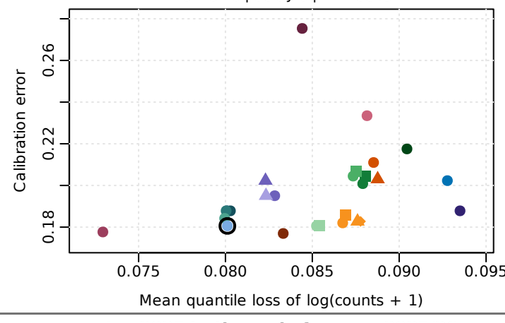
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 0 | 68 weeks



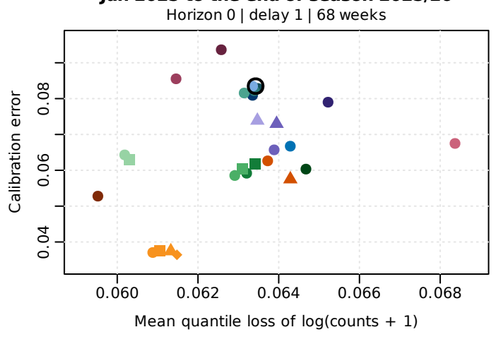
Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 19 weeks



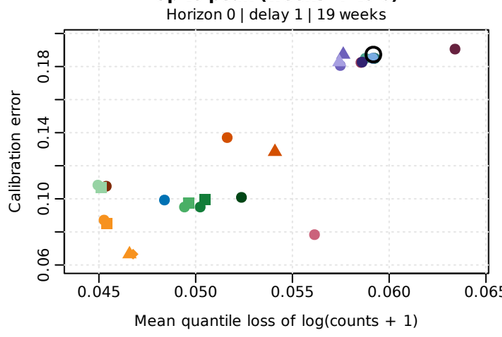
After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 12 weeks



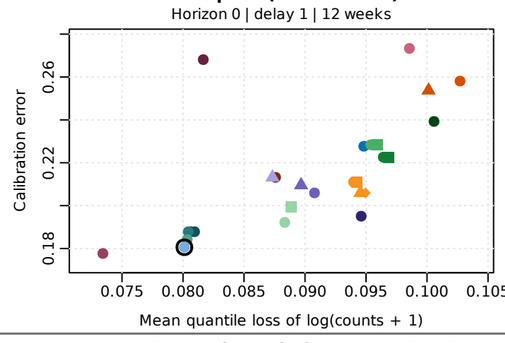
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 1 | 68 weeks



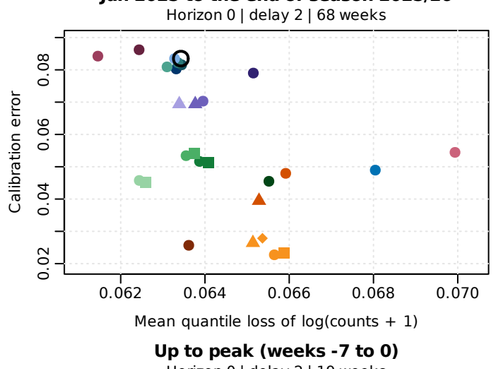
Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 19 weeks



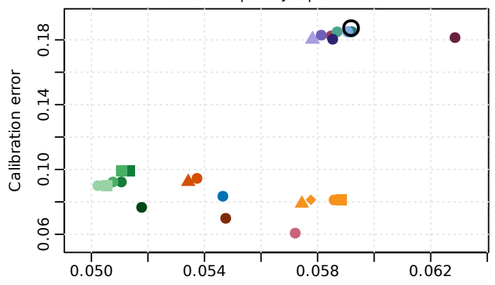
After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 12 weeks



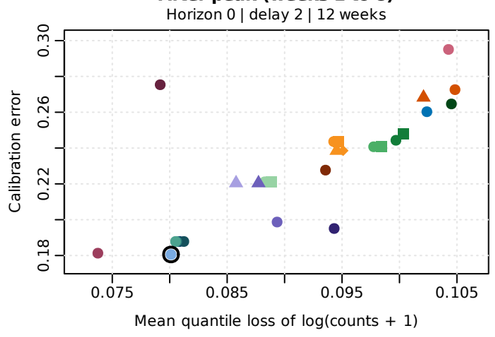
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 2 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 2 | 19 weeks

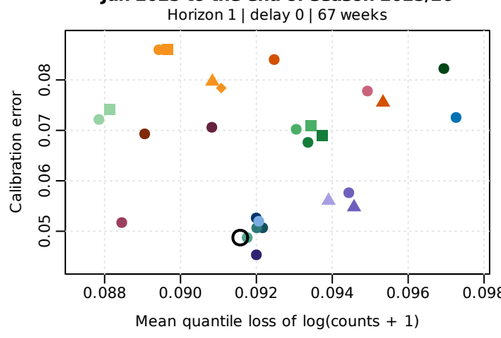


After peak (weeks 1 to 8)
Horizon 0 | delay 2 | 12 weeks

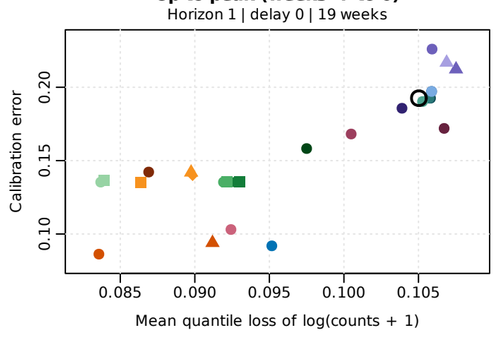


Horizon 1

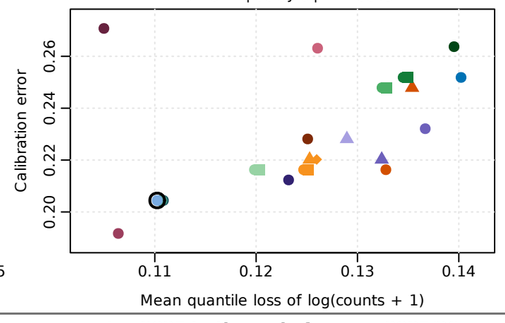
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 0 | 67 weeks



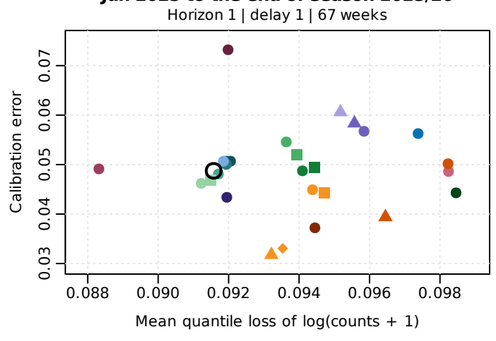
Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 19 weeks



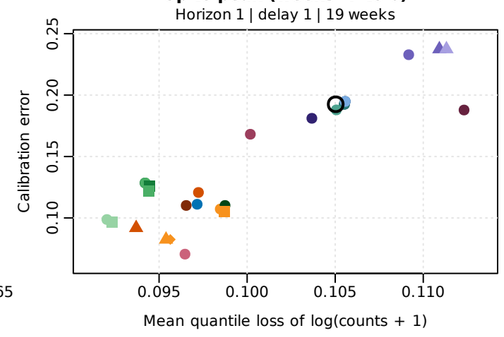
After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 11 weeks



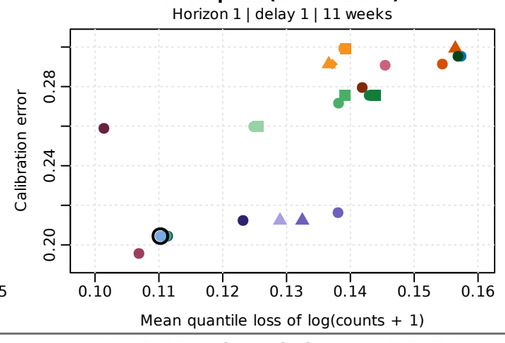
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 1 | 67 weeks



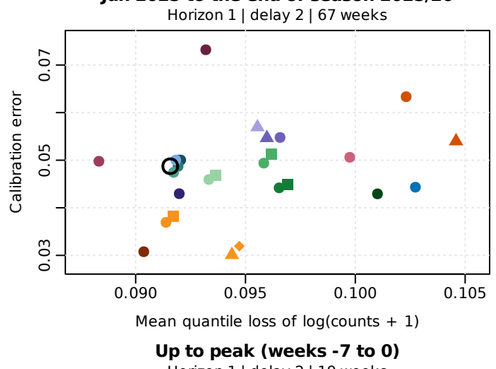
Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 19 weeks



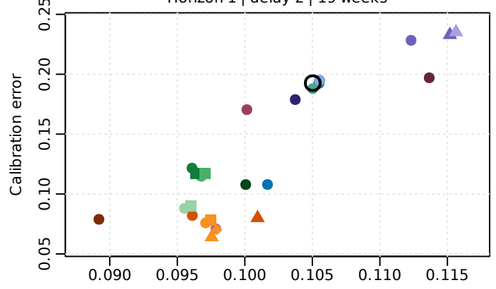
After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 11 weeks



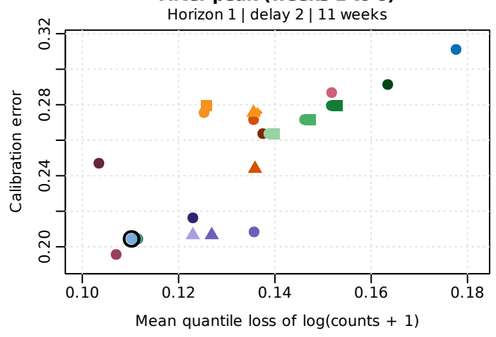
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 2 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 2 | 19 weeks

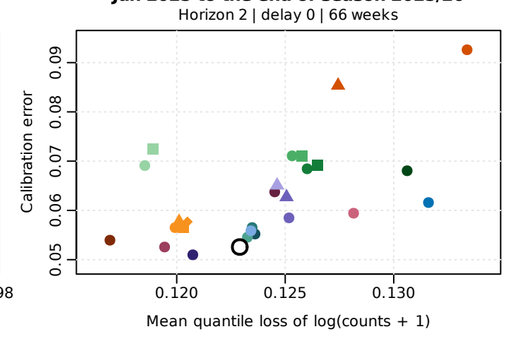


After peak (weeks 1 to 8)
Horizon 1 | delay 2 | 11 weeks

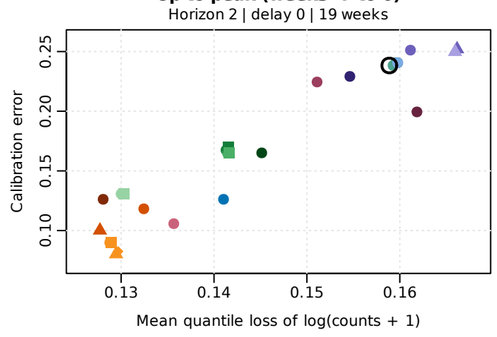


Horizon 2

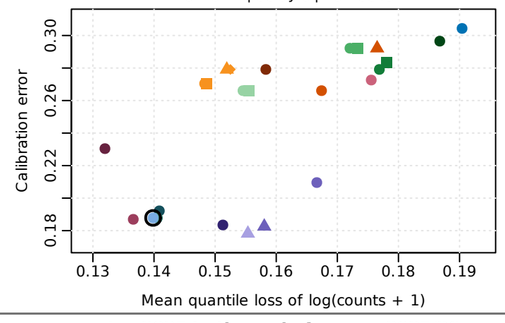
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 0 | 66 weeks



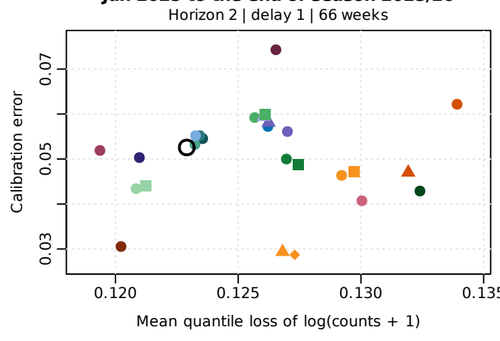
Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 19 weeks



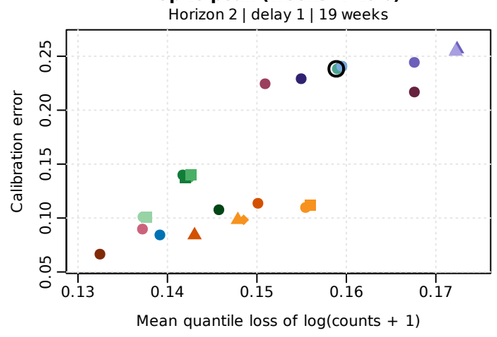
After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 10 weeks



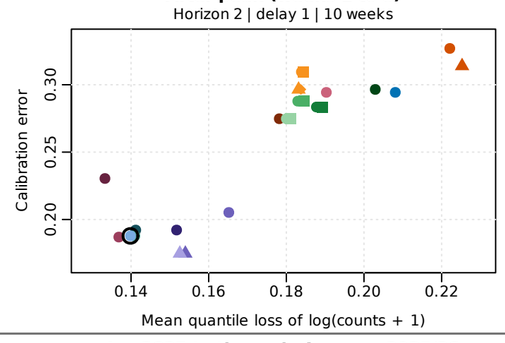
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 1 | 66 weeks



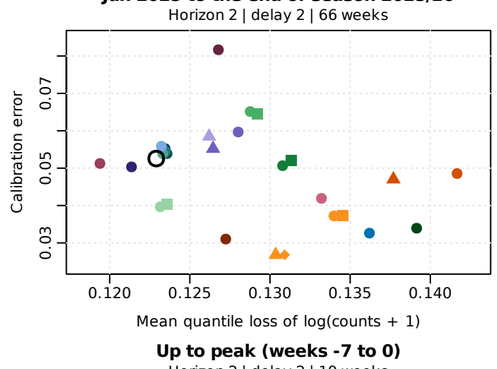
Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 19 weeks



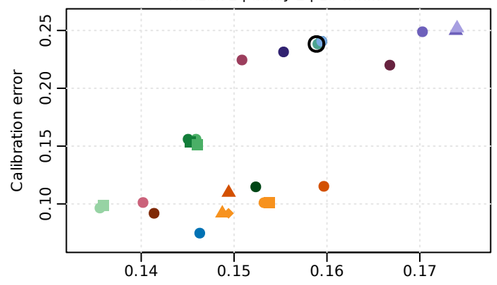
After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 10 weeks



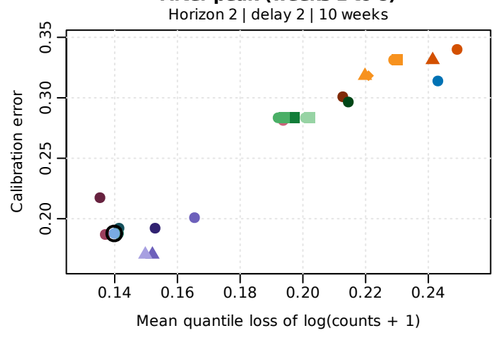
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 2 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 2 | 19 weeks

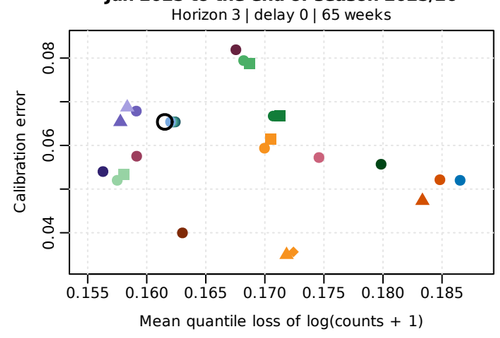


After peak (weeks 1 to 8)
Horizon 2 | delay 2 | 10 weeks

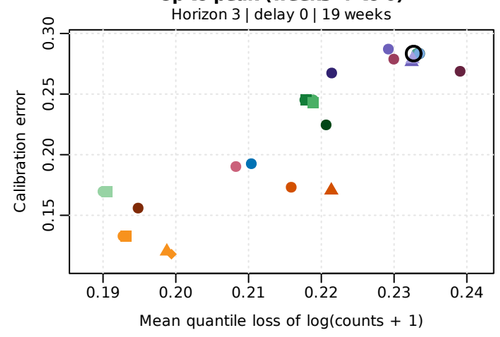


Horizon 3

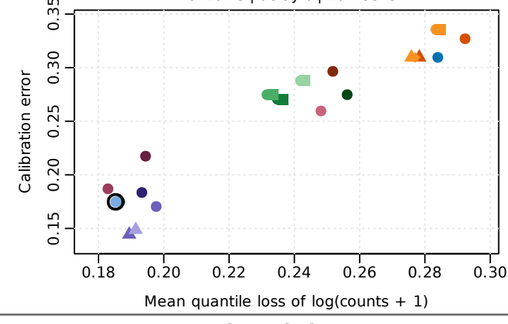
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 0 | 65 weeks



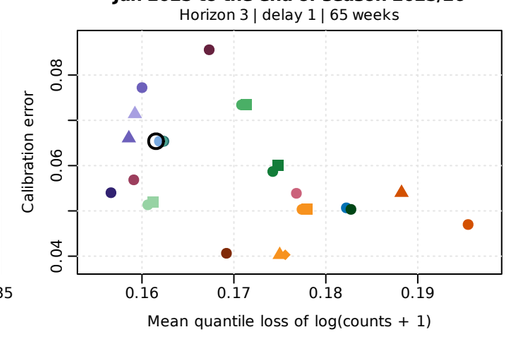
Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 19 weeks



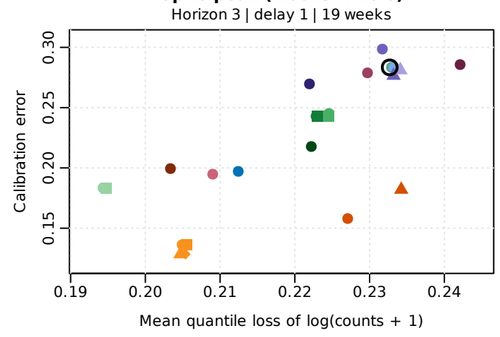
After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 10 weeks



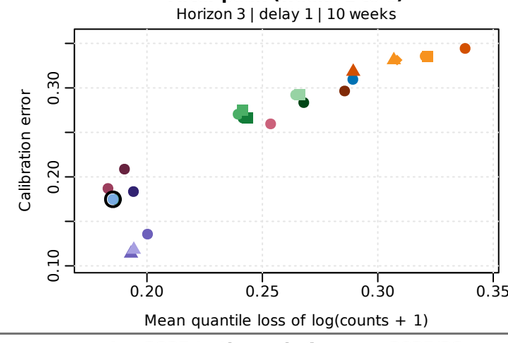
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 1 | 65 weeks



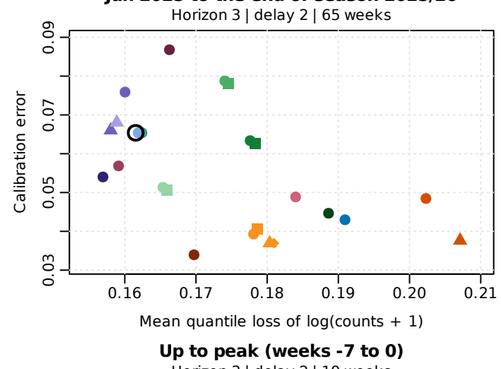
Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 19 weeks



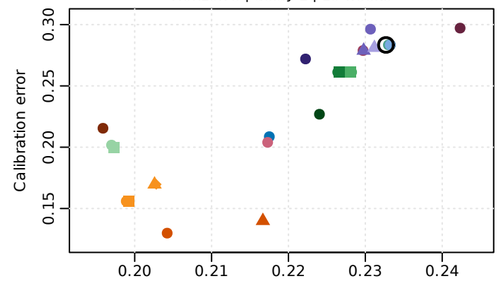
After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 10 weeks



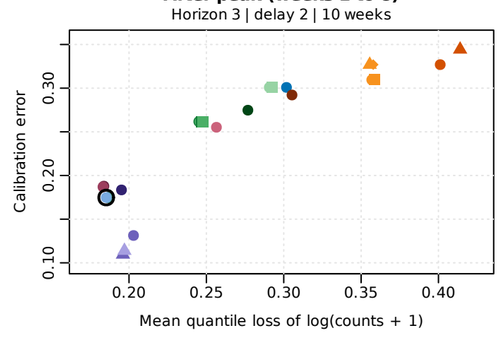
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 2 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 2 | 19 weeks



After peak (weeks 1 to 8)
Horizon 3 | delay 2 | 10 weeks



Feedback delay 0 week(s)

Feedback delay 1 week(s)

Feedback delay 2 week(s)

Single trackers
- Original
- Fixed MultiQT (eta=0.001*S0)
- Log-slope rate (min m=0.002)
- Target oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=2)
- Rate grid: learned + sharing, common weights (H=2)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Diverse experts
- Diverse experts: equal (slow m=0.001)
- Diverse experts: learned + sharing (slow m=0.001, H=8)
- Diverse experts: learned + sharing, common weights (slow m=0.004)
- Diverse experts: soft switching (strength=1)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.003)
- Studentized median (base 80% width, normal start, eta=0.003)
- Median, constant sigma (RMS of 5, normal start, eta=0.1)

Studentized full base
- Studentized full base (base 80% width, eta=0.003)
- Studentized full base (level scales, eta=0.003)
- Studentized full base (level RMS of 5, eta=0.003)

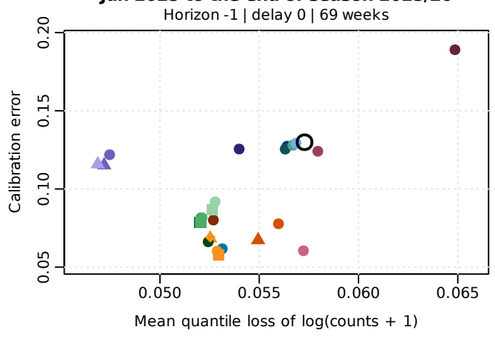
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | New York (36) | evaluation: Jun 2025 to the end of season 2025/26 | log1p(counts) updates, loss of log(counts + 1)

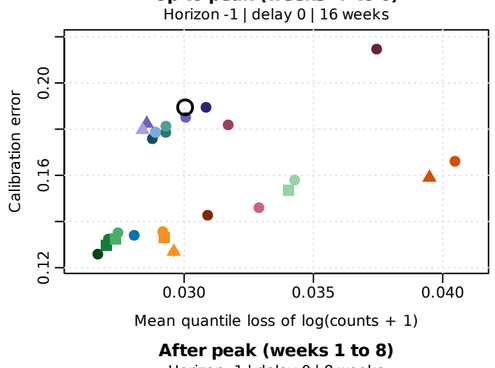
Rows: feedback delay (MultiQT learns from the report that many weeks after the first one) and window; columns: horizon. Every delay uses the settings tuned with feedback delay 0.

Horizon -1

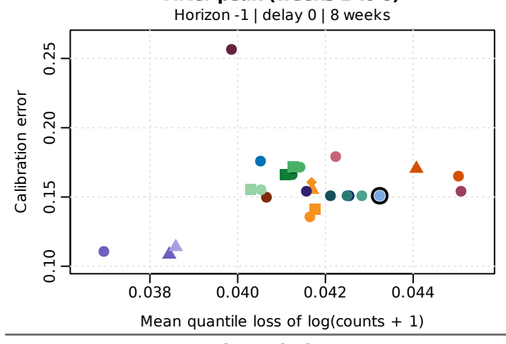
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 0 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 16 weeks

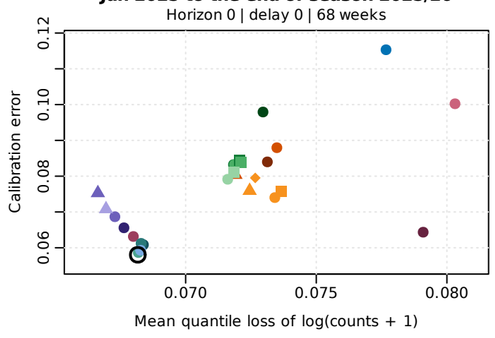


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 8 weeks

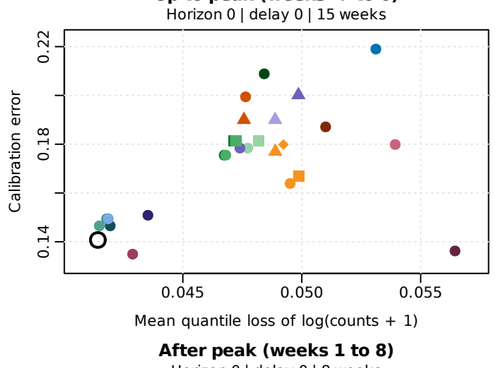


Horizon 0

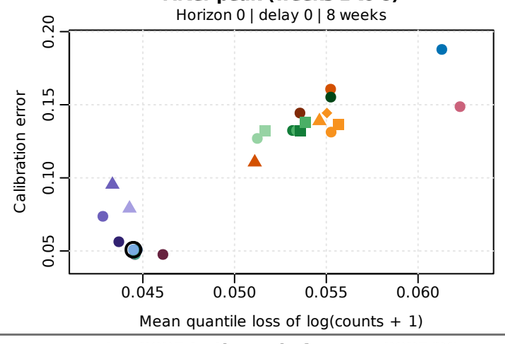
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 0 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 15 weeks

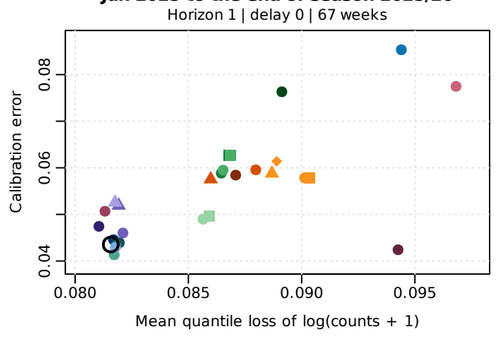


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 8 weeks

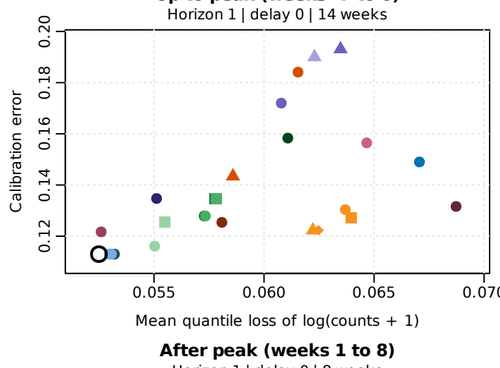


Horizon 1

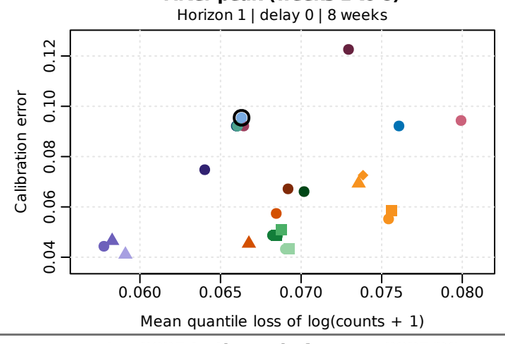
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 0 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 14 weeks

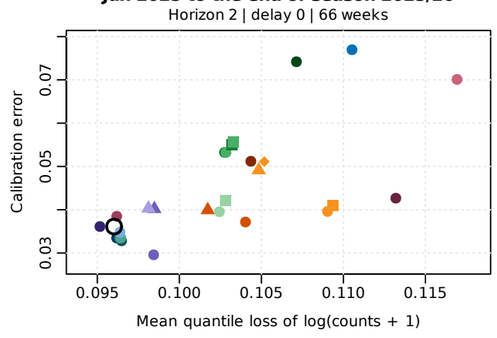


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 8 weeks

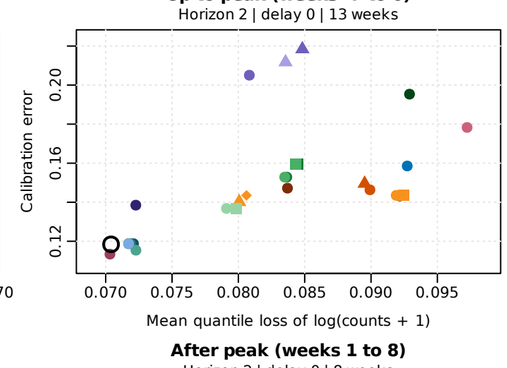


Horizon 2

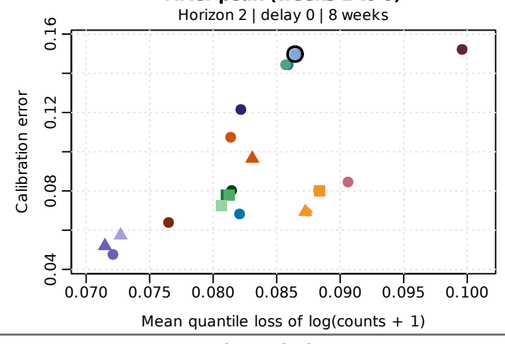
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 0 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 13 weeks

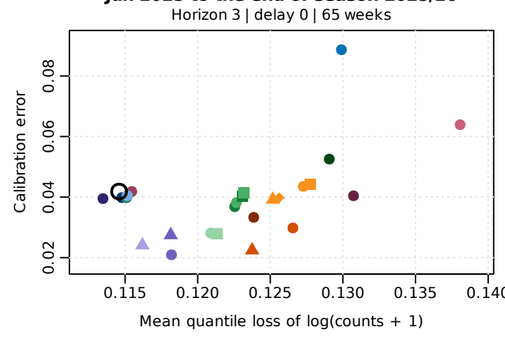


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 8 weeks

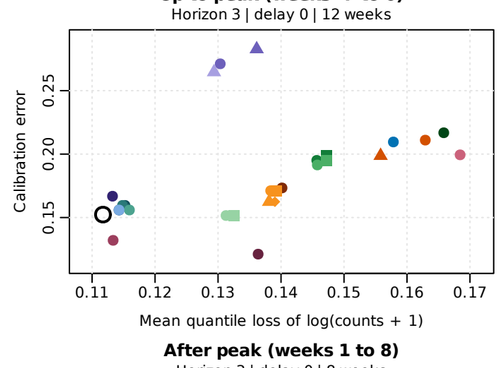


Horizon 3

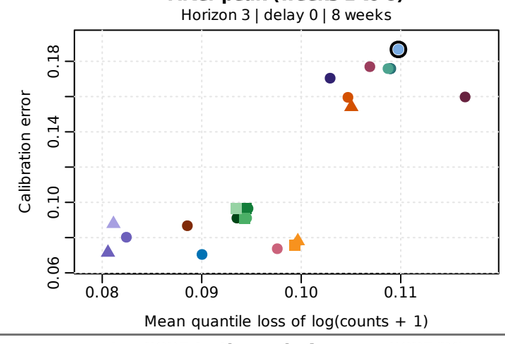
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 0 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 12 weeks



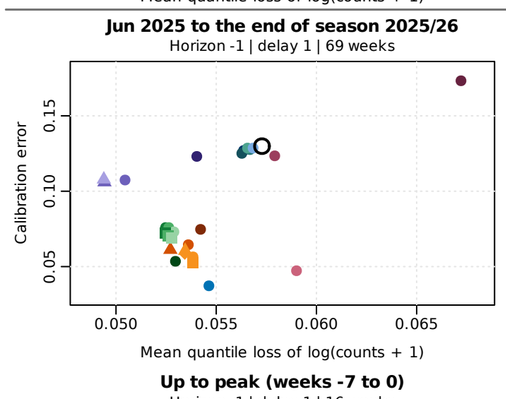
After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 8 weeks



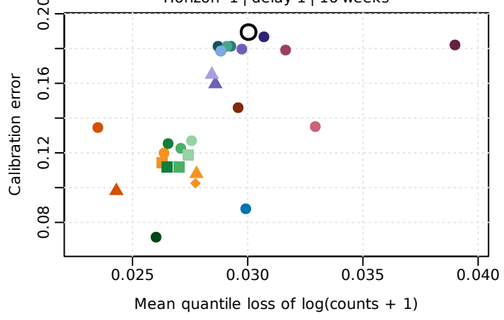
Feedback delay 0 week(s)

Horizon -1

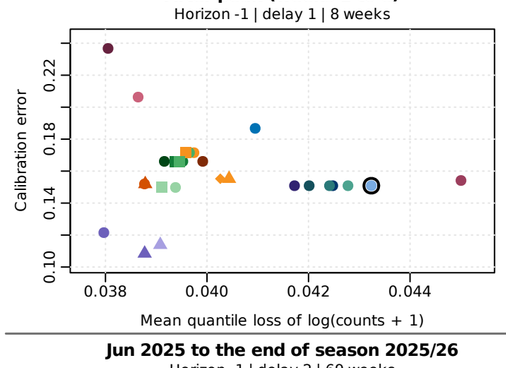
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 1 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 16 weeks

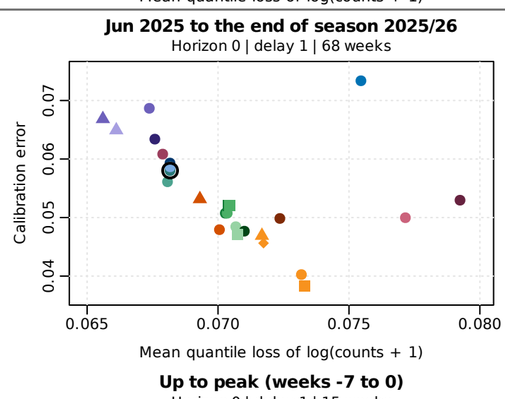


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 8 weeks

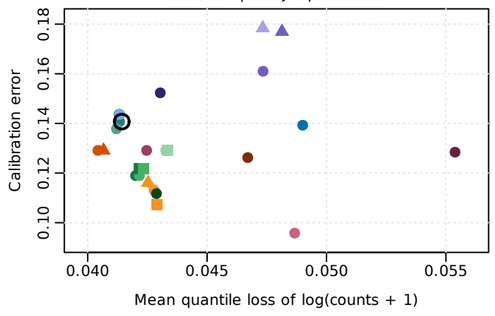


Horizon 0

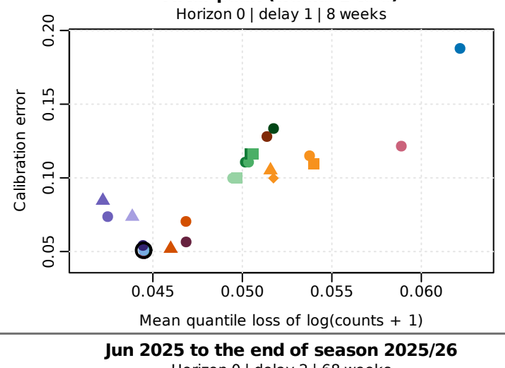
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 1 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 15 weeks

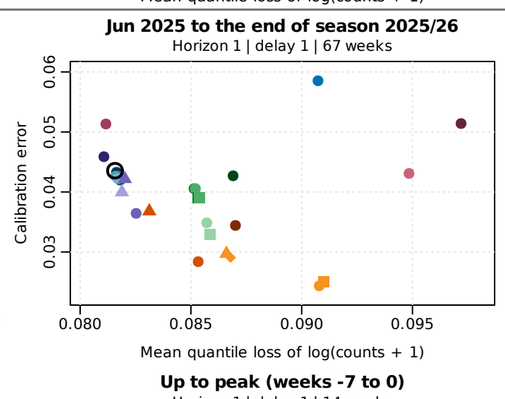


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 8 weeks

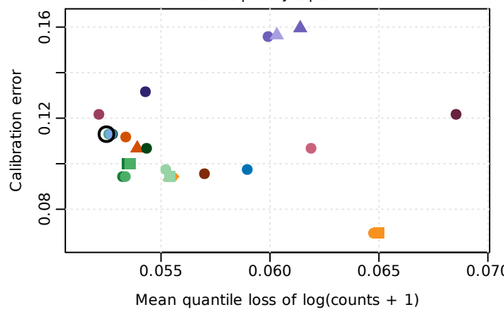


Horizon 1

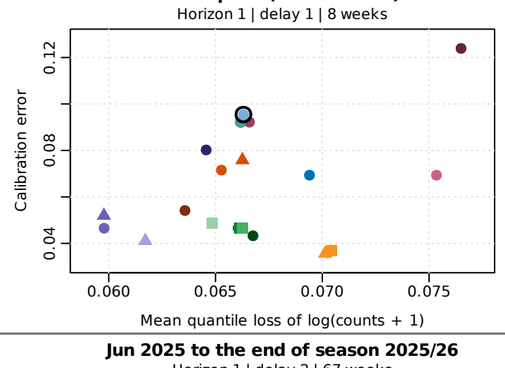
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 1 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 14 weeks

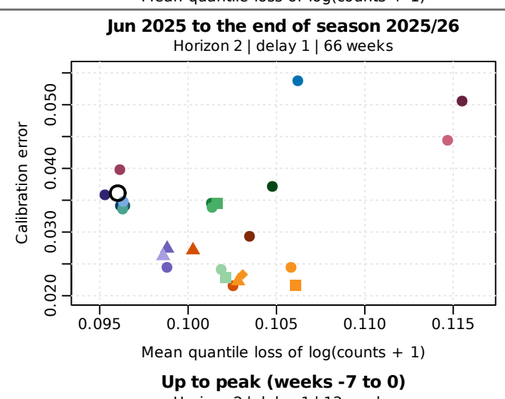


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 8 weeks

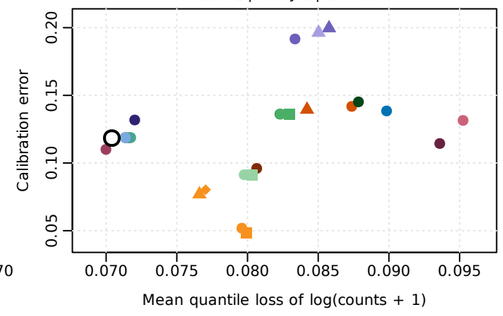


Horizon 2

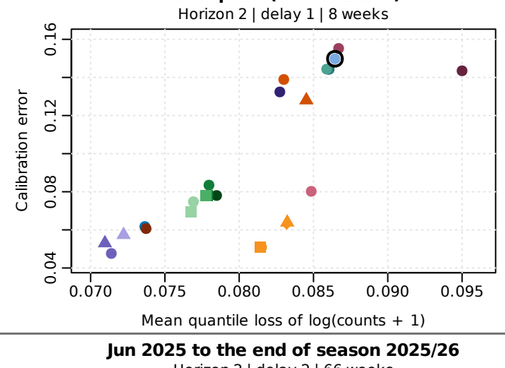
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 1 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 13 weeks

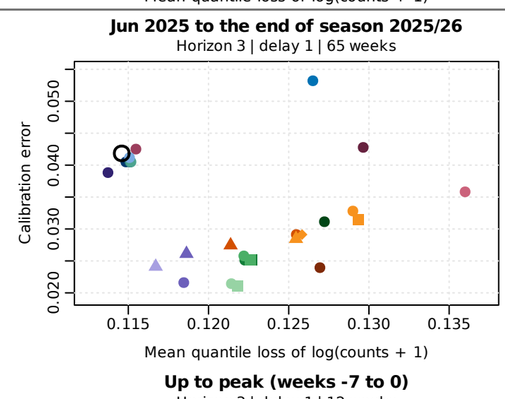


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 8 weeks

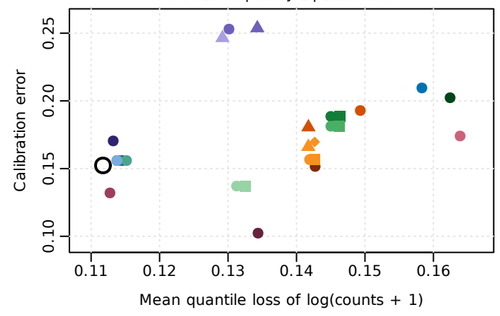


Horizon 3

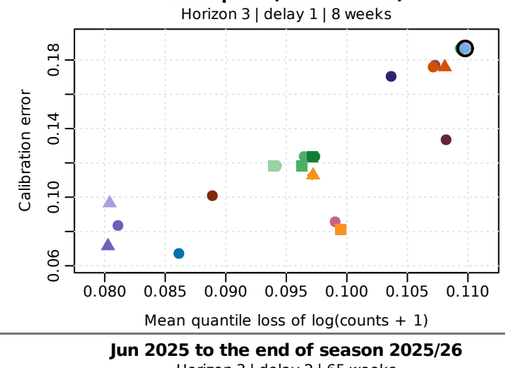
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 1 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 12 weeks



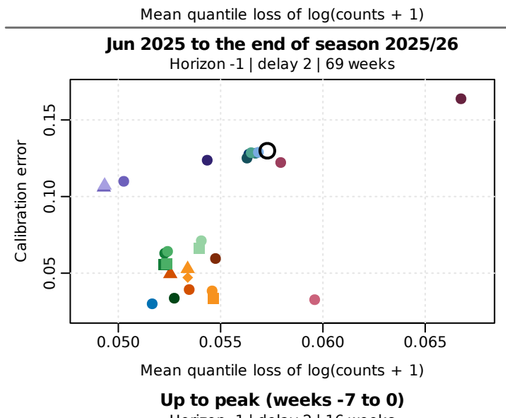
After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 8 weeks



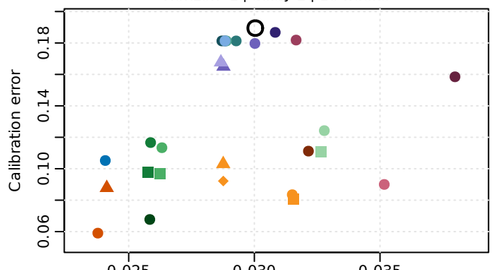
Feedback delay 1 week(s)

Horizon -1

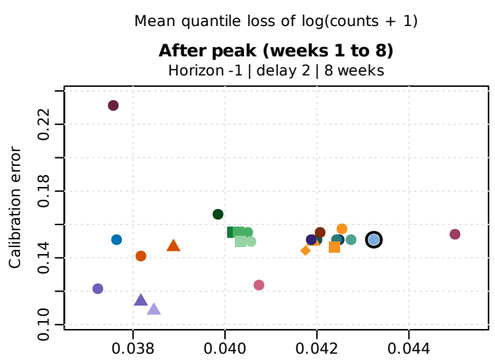
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 2 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 2 | 16 weeks

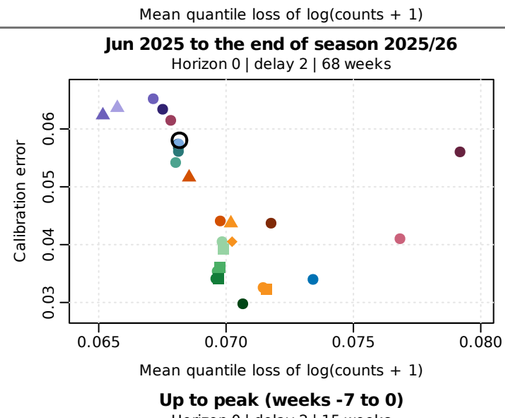


After peak (weeks 1 to 8)
Horizon -1 | delay 2 | 8 weeks

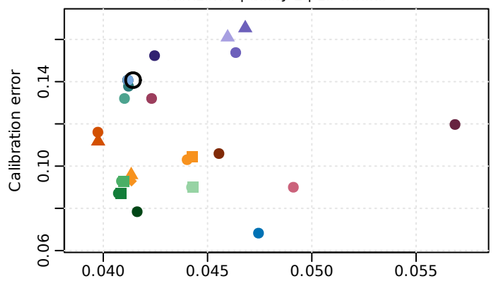


Horizon 0

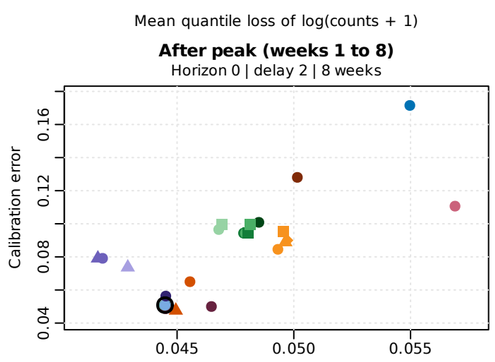
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 2 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 2 | 15 weeks

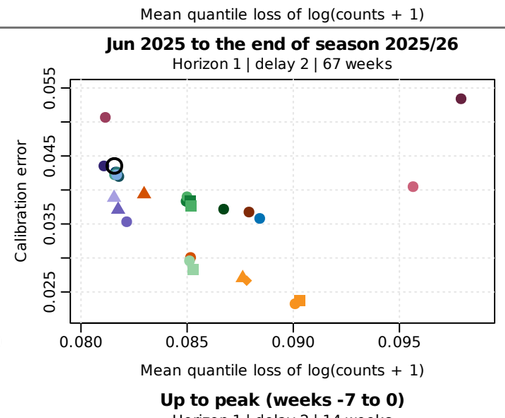


After peak (weeks 1 to 8)
Horizon 0 | delay 2 | 8 weeks

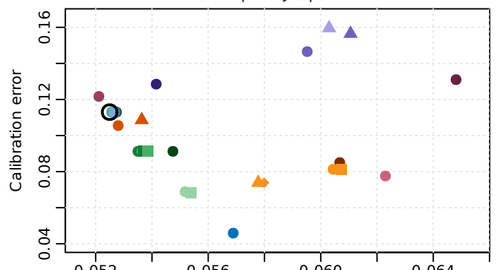


Horizon 1

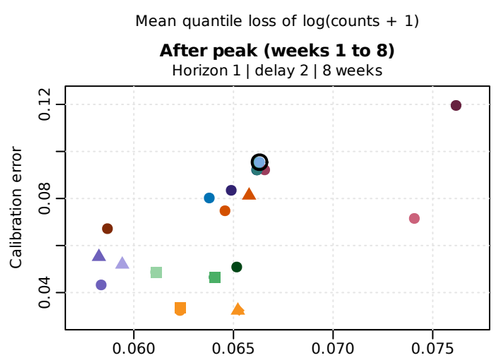
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 2 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 2 | 14 weeks

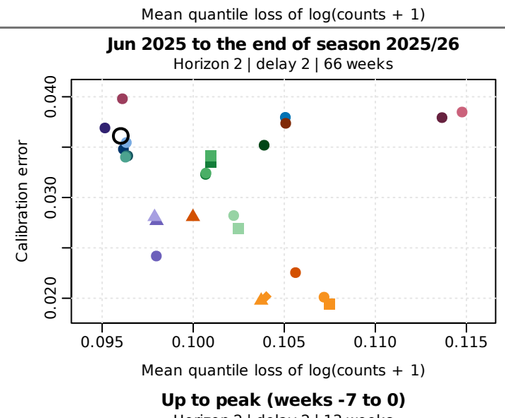


After peak (weeks 1 to 8)
Horizon 1 | delay 2 | 8 weeks

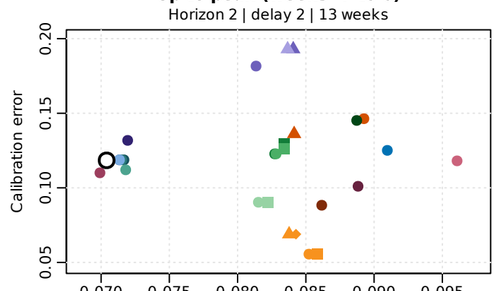


Horizon 2

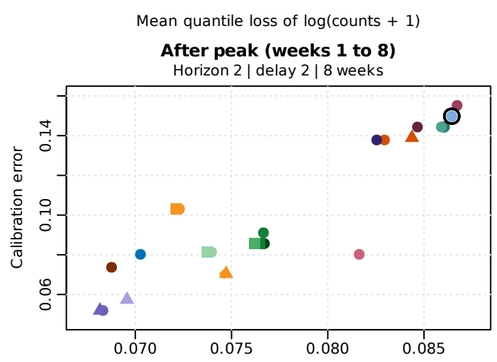
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 2 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 2 | 13 weeks

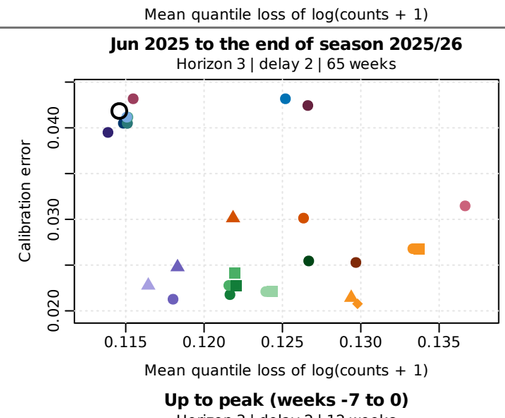


After peak (weeks 1 to 8)
Horizon 2 | delay 2 | 8 weeks

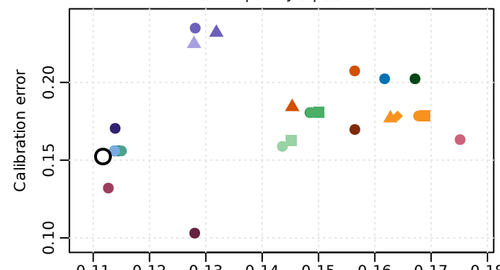


Horizon 3

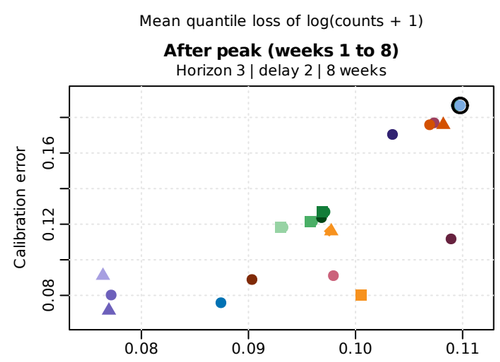
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 2 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 2 | 12 weeks



After peak (weeks 1 to 8)
Horizon 3 | delay 2 | 8 weeks



Feedback delay 2 week(s)

Single trackers
- Original
- Fixed MultiQT (eta=0.001*S0)
- Log-slope rate (min m=0.002)
- Target oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=2)
- Rate grid: learned + sharing, common weights (H=2)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Diverse experts
- Diverse experts: equal (slow m=0.001)
- Diverse experts: learned + sharing (slow m=0.001, H=8)
- Diverse experts: learned + sharing, common weights (slow m=0.004)
- Diverse experts: soft switching (strength=1)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.003)
- Studentized median (base 80% width, normal start, eta=0.003)
- Median, constant sigma (RMS of 5, normal start, eta=0.1)

Studentized full base
- Studentized full base (base 80% width, eta=0.003)
- Studentized full base (level scales, eta=0.003)
- Studentized full base (level RMS of 5, eta=0.003)

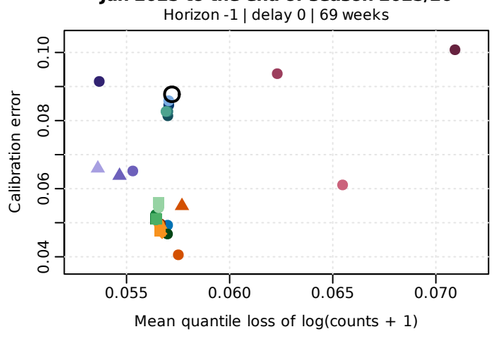
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

covid | Pennsylvania (42) | evaluation: Jun 2025 to the end of season 2025/26 | log1p(counts) updates, loss of log(counts + 1)

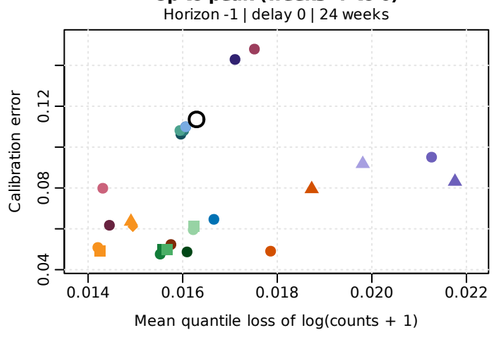
Rows: feedback delay (MultiQT learns from the report that many weeks after the first one) and window; columns: horizon. Every delay uses the settings tuned with feedback delay 0.

Horizon -1

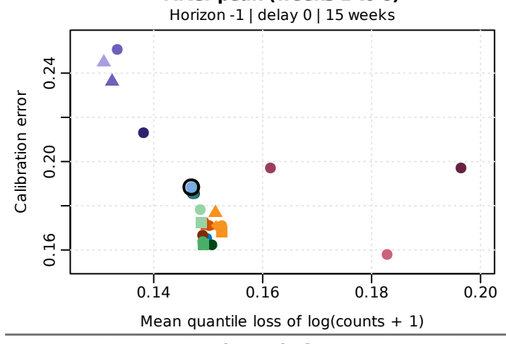
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 0 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 24 weeks

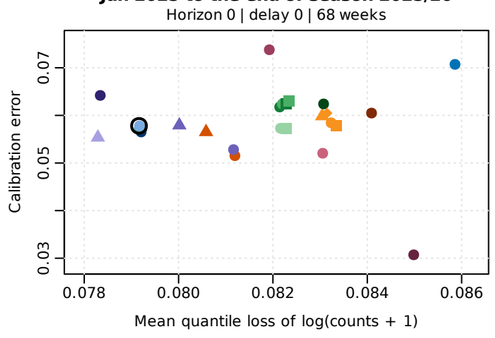


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 15 weeks

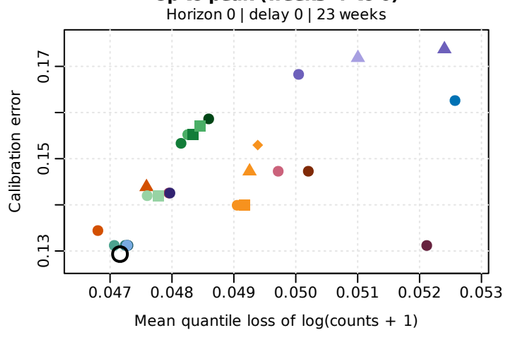


Horizon 0

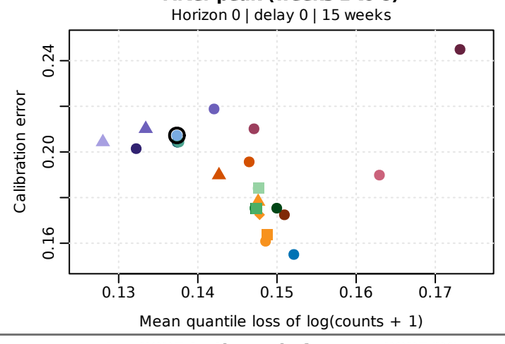
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 0 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 23 weeks

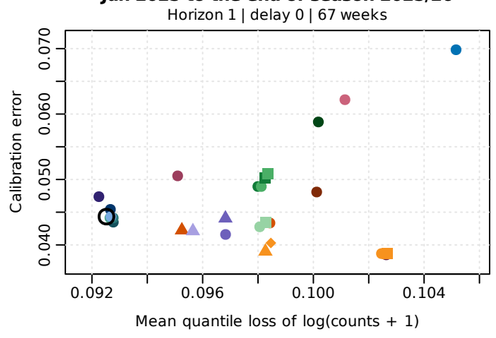


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 15 weeks

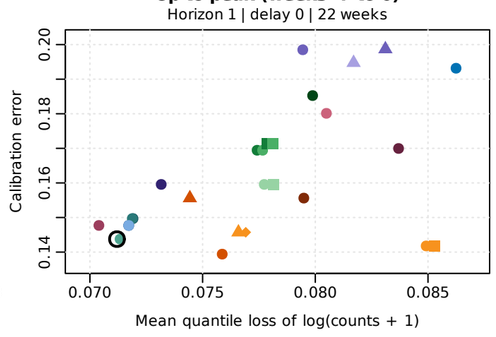


Horizon 1

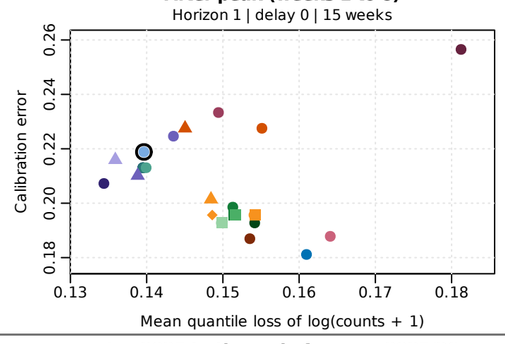
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 0 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 22 weeks

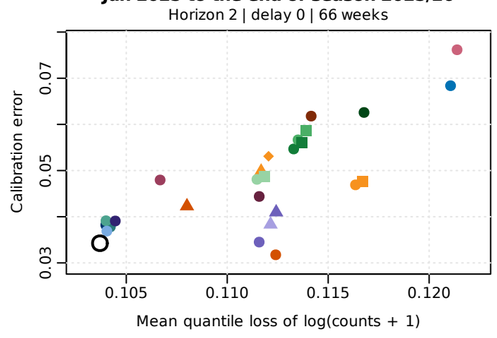


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 15 weeks

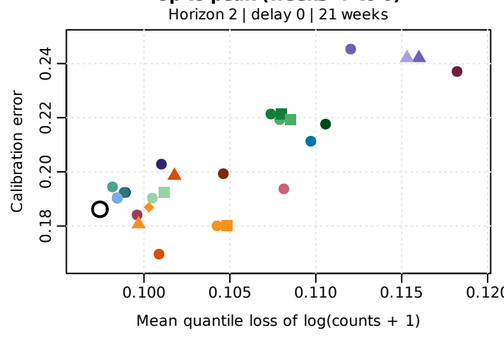


Horizon 2

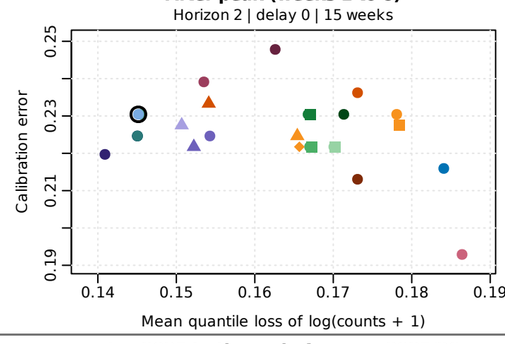
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 0 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 21 weeks

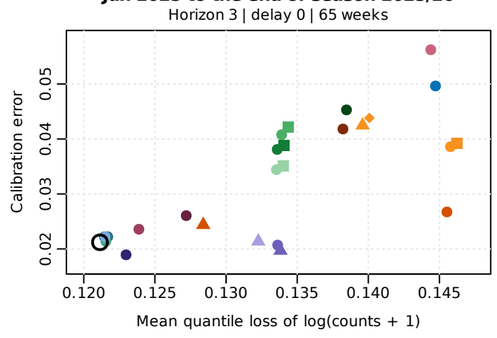


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 15 weeks

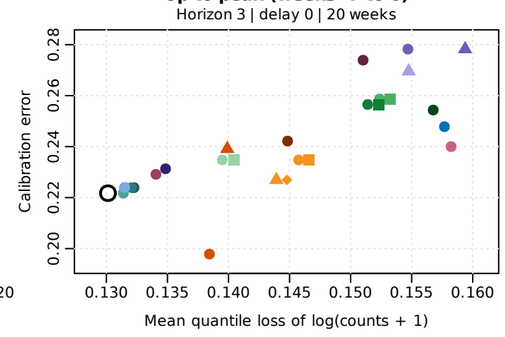


Horizon 3

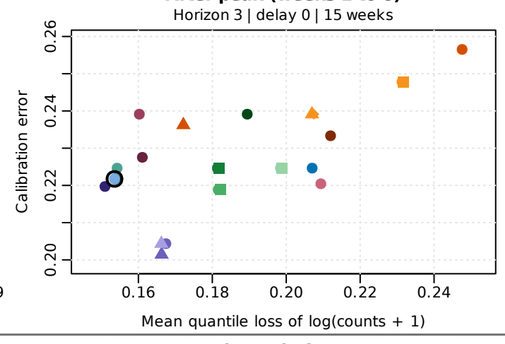
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 0 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 20 weeks



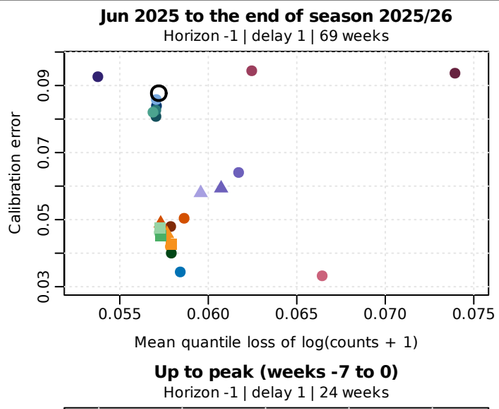
After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 15 weeks



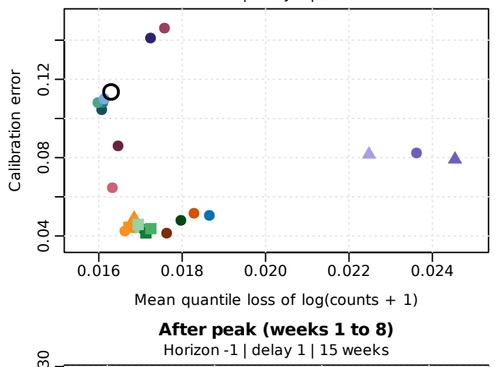
Feedback delay 1 week(s)

Horizon -1

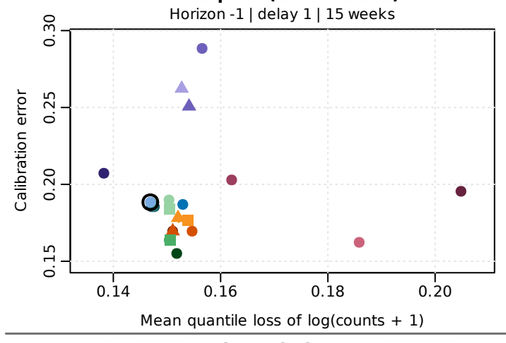
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 1 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 24 weeks

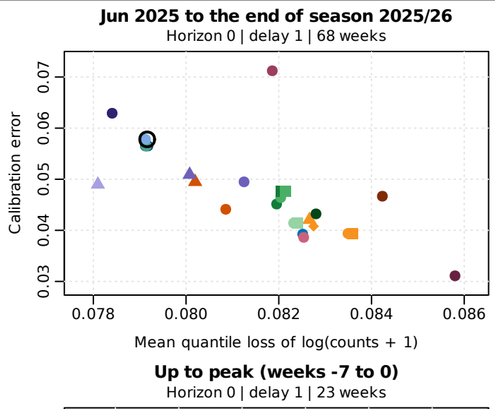


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 15 weeks

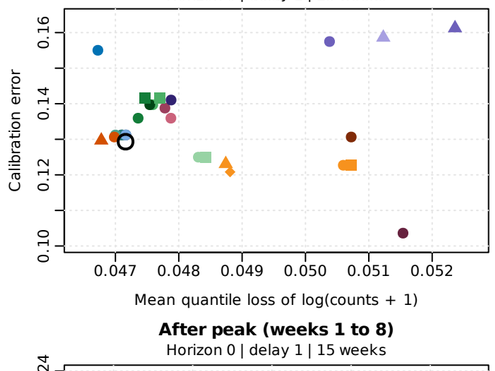


Horizon 0

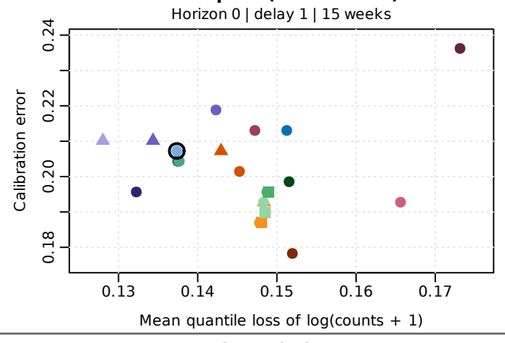
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 1 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 23 weeks

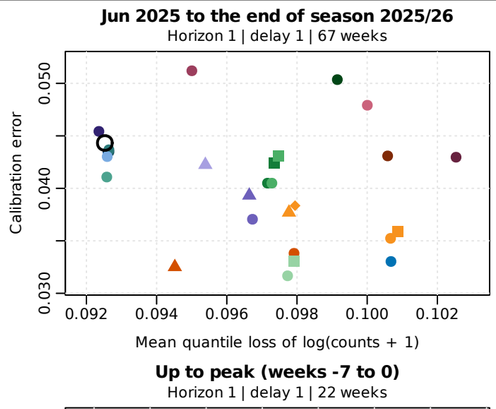


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 15 weeks

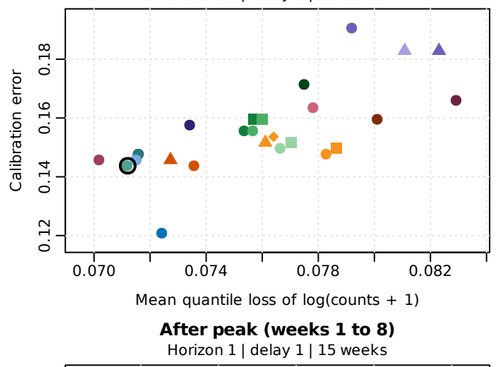


Horizon 1

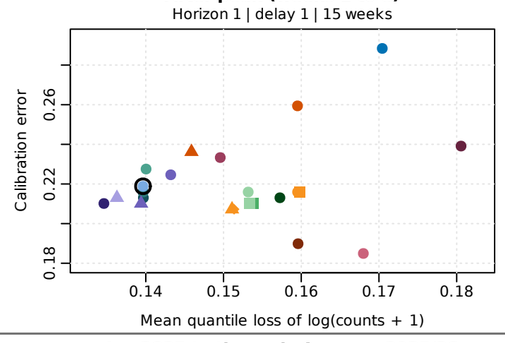
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 1 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 22 weeks

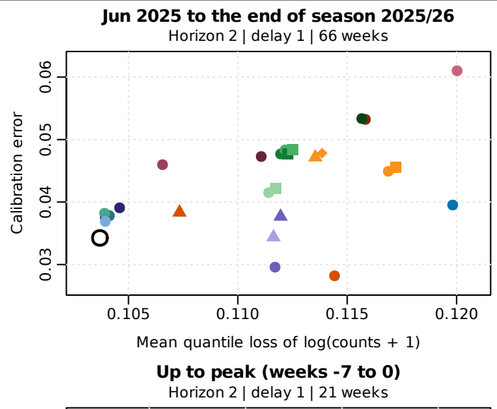


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 15 weeks

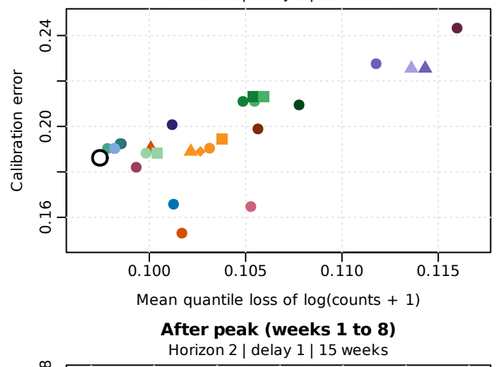


Horizon 2

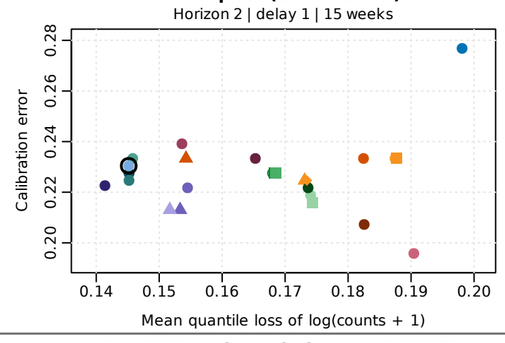
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 1 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 21 weeks

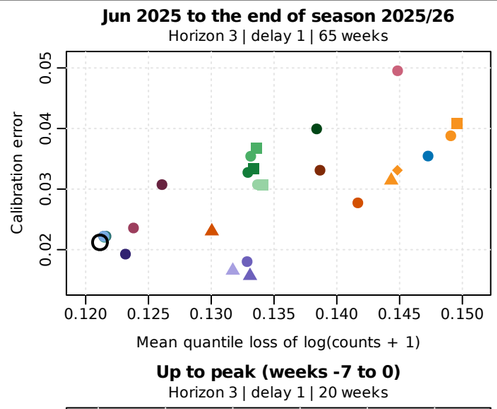


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 15 weeks

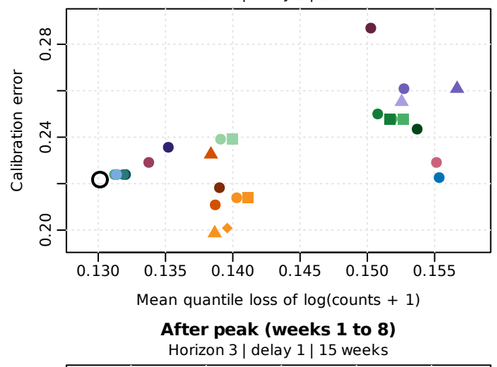


Horizon 3

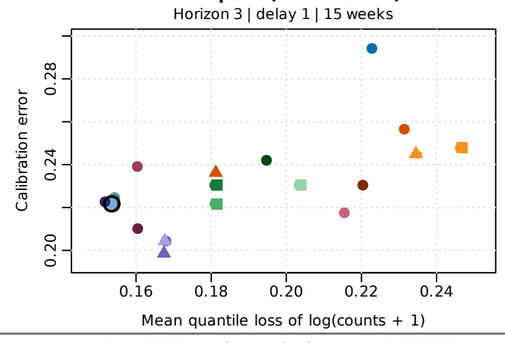
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 1 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 20 weeks



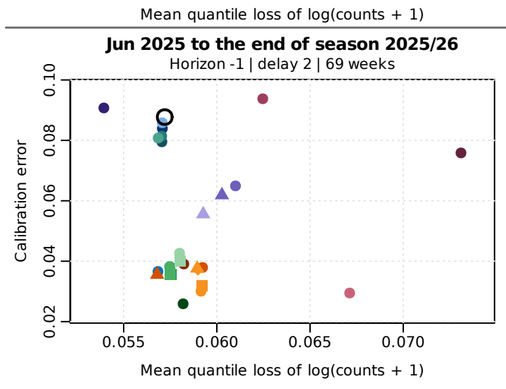
After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 15 weeks



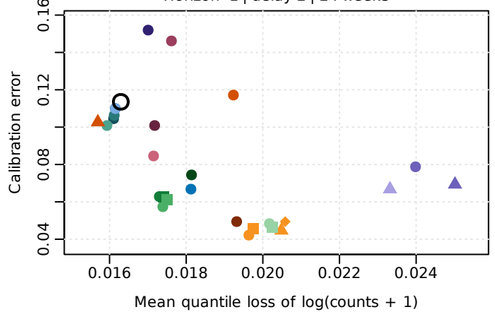
Feedback delay 2 week(s)

Horizon -1

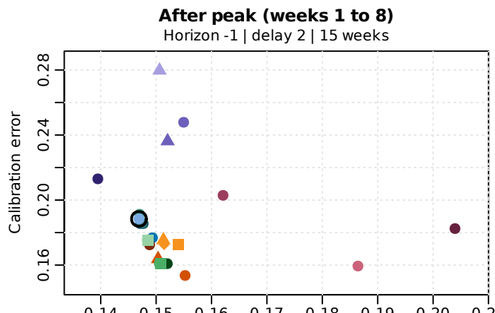
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 2 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 2 | 24 weeks

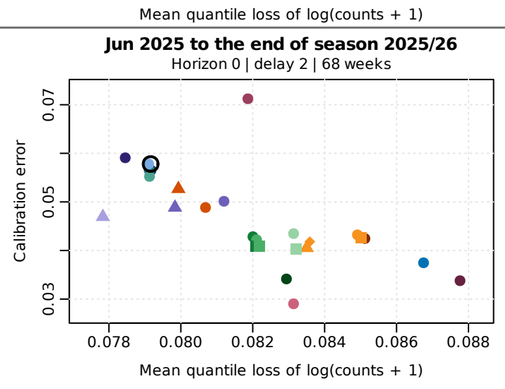


After peak (weeks 1 to 8)
Horizon -1 | delay 2 | 15 weeks

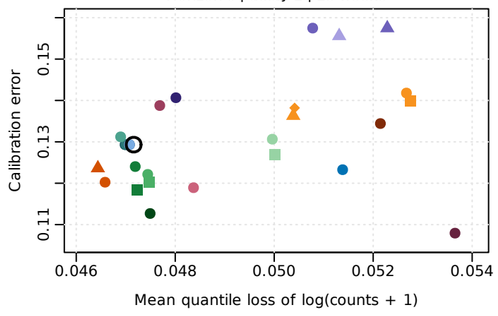


Horizon 0

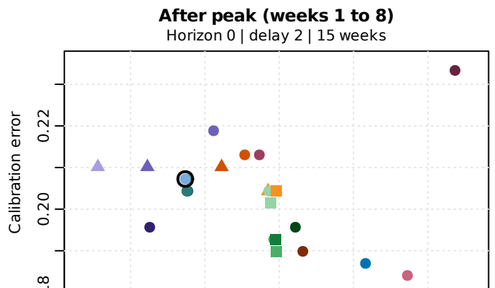
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 2 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 2 | 23 weeks

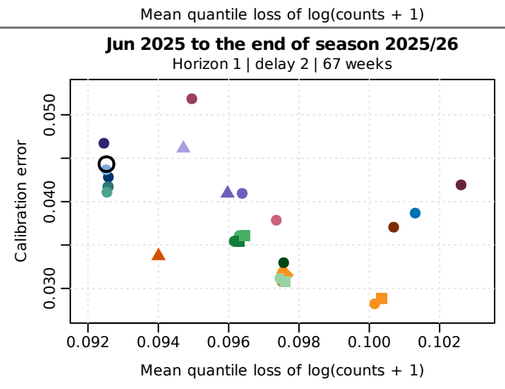


After peak (weeks 1 to 8)
Horizon 0 | delay 2 | 15 weeks

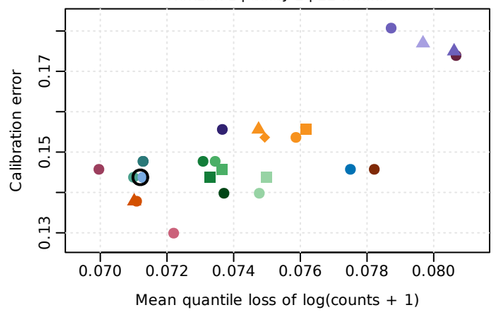


Horizon 1

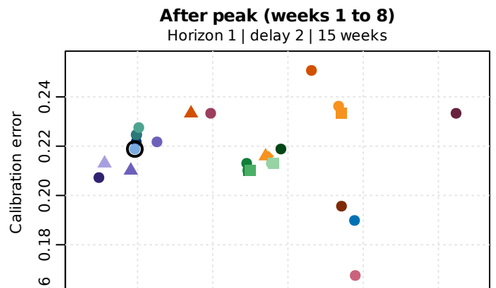
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 2 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 2 | 22 weeks

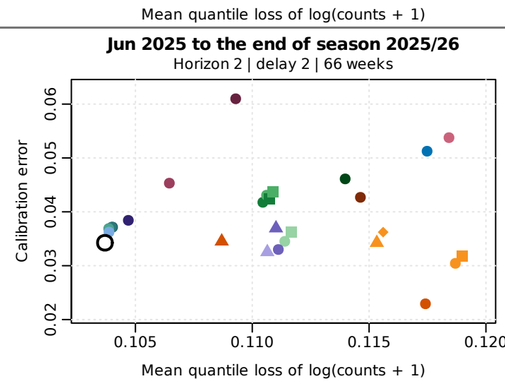


After peak (weeks 1 to 8)
Horizon 1 | delay 2 | 15 weeks

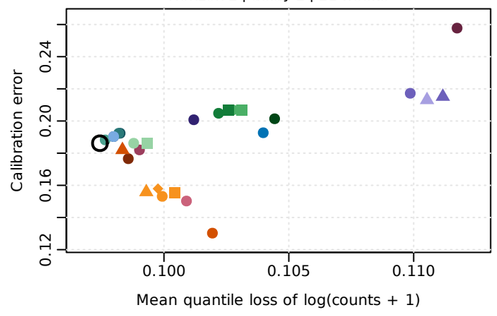


Horizon 2

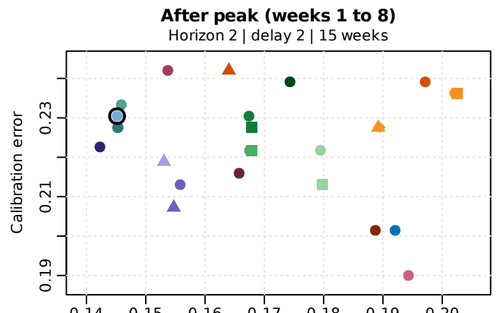
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 2 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 2 | 21 weeks

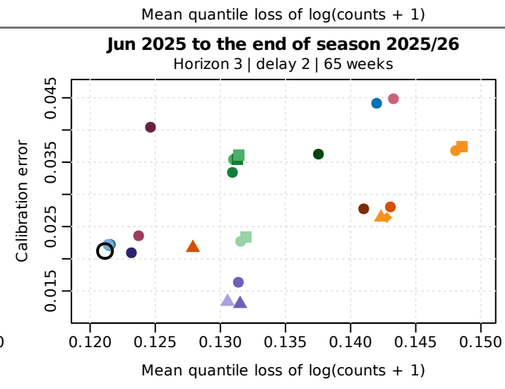


After peak (weeks 1 to 8)
Horizon 2 | delay 2 | 15 weeks

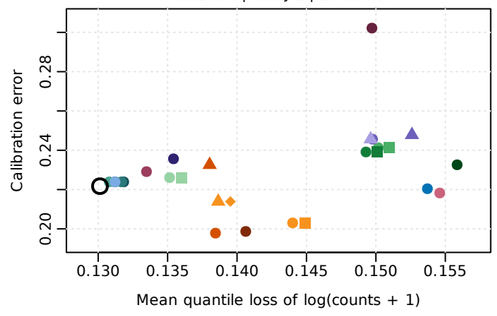


Horizon 3

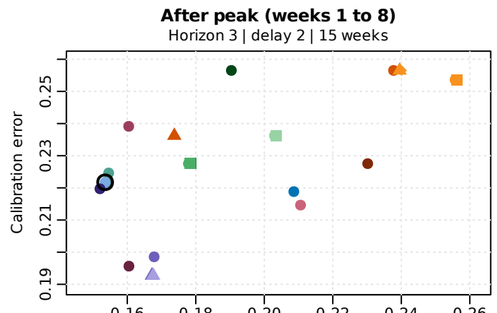
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 2 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 2 | 20 weeks



After peak (weeks 1 to 8)
Horizon 3 | delay 2 | 15 weeks



Single trackers
- Original
- Fixed MultiQT (eta=0.001*S0)
- Log-slope rate (min m=0.002)
- Target oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=2)
- Rate grid: learned + sharing, common weights (H=2)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Diverse experts
- Diverse experts: equal (slow m=0.001)
- Diverse experts: learned + sharing (slow m=0.001, H=8)
- Diverse experts: learned + sharing, common weights (slow m=0.004)
- Diverse experts: soft switching (strength=1)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.003)
- Studentized median (base 80% width, normal start, eta=0.003)
- Median, constant sigma (RMS of 5, normal start, eta=0.1)

Studentized full base
- Studentized full base (base 80% width, eta=0.003)
- Studentized full base (level scales, eta=0.003)
- Studentized full base (level RMS of 5, eta=0.003)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).

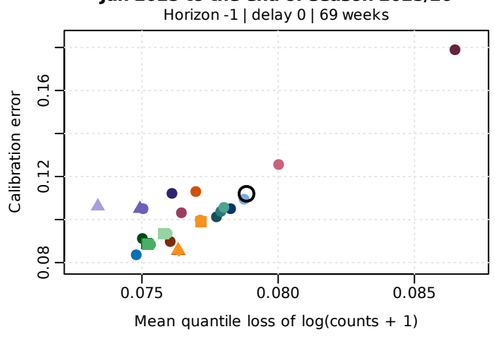
covid | Illinois (17) | evaluation: Jun 2025 to the end of season 2025/26 | log1p(counts) updates, loss of log(counts + 1)

Rows: feedback delay (MultiQT learns from the report that many weeks after the first one) and window; columns: horizon. Every delay uses the settings tuned with feedback delay 0.

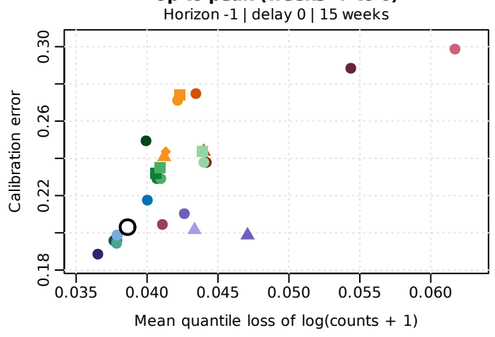
Horizon -1

Feedback delay 0 week(s)

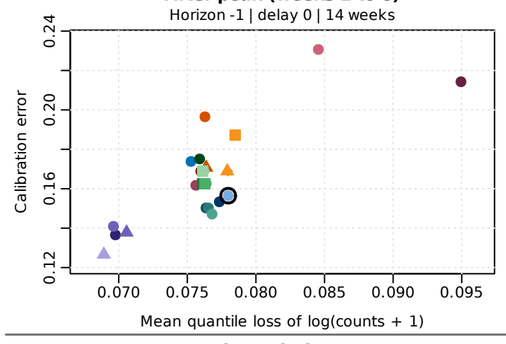
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 0 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 0 | 15 weeks

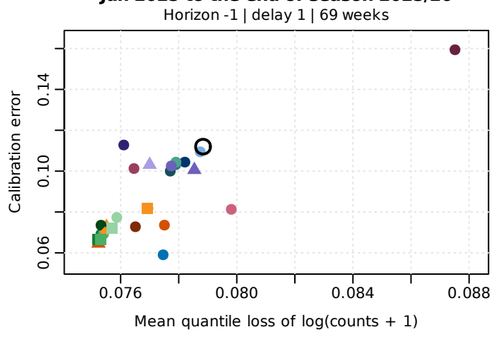


After peak (weeks 1 to 8)
Horizon -1 | delay 0 | 14 weeks

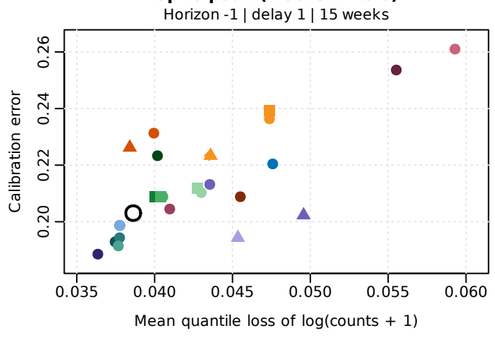


Feedback delay 1 week(s)

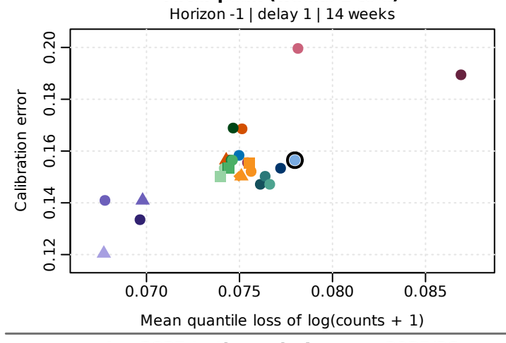
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 1 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 1 | 15 weeks

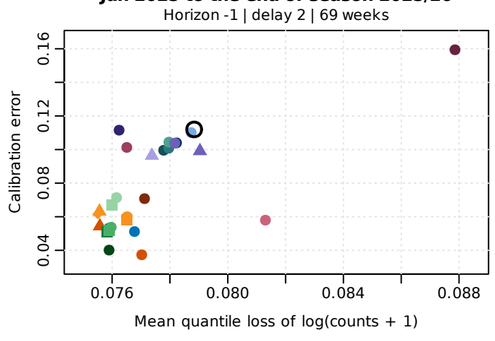


After peak (weeks 1 to 8)
Horizon -1 | delay 1 | 14 weeks

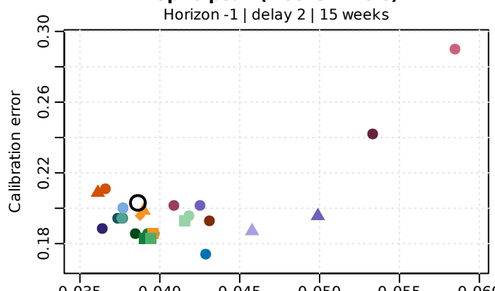


Feedback delay 2 week(s)

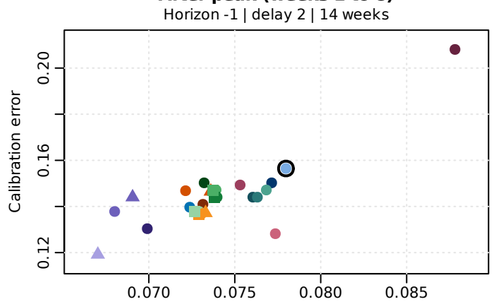
Jun 2025 to the end of season 2025/26
Horizon -1 | delay 2 | 69 weeks



Up to peak (weeks -7 to 0)
Horizon -1 | delay 2 | 15 weeks



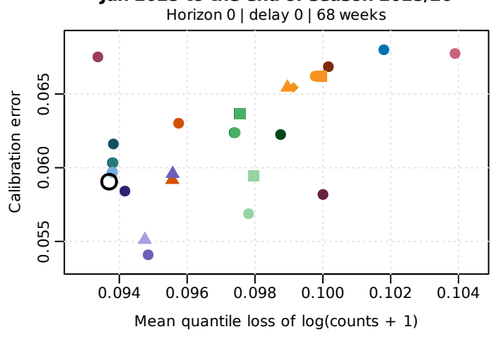
After peak (weeks 1 to 8)
Horizon -1 | delay 2 | 14 weeks



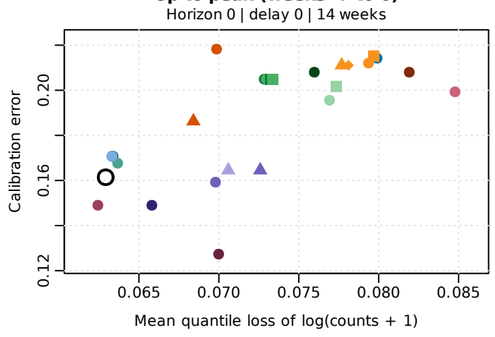
Horizon 0

Feedback delay 0 week(s)

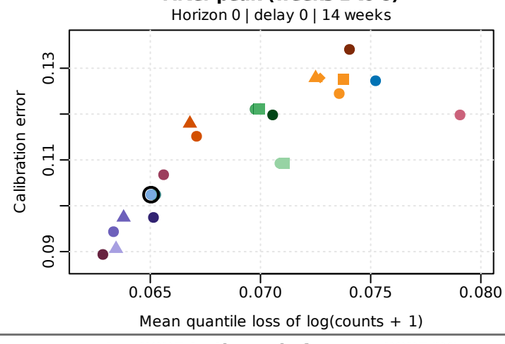
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 0 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 0 | 14 weeks

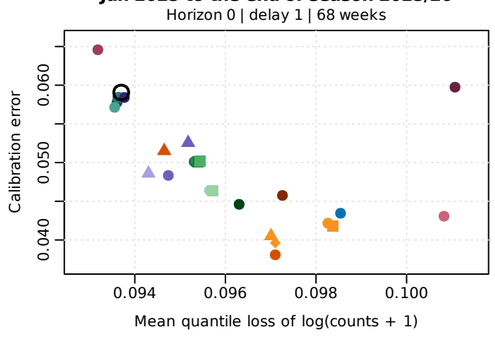


After peak (weeks 1 to 8)
Horizon 0 | delay 0 | 14 weeks

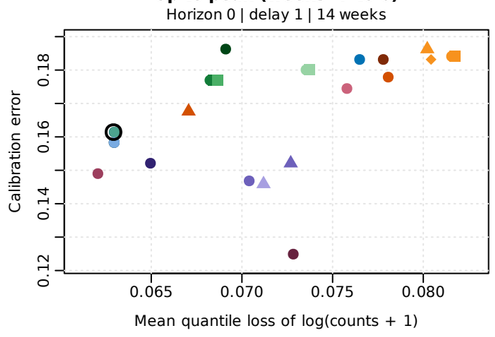


Feedback delay 1 week(s)

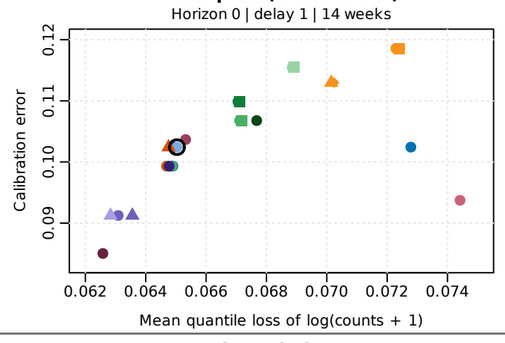
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 1 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 1 | 14 weeks

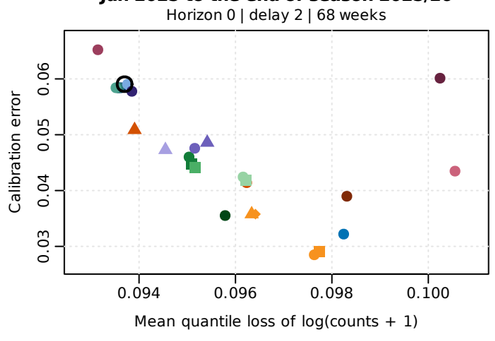


After peak (weeks 1 to 8)
Horizon 0 | delay 1 | 14 weeks

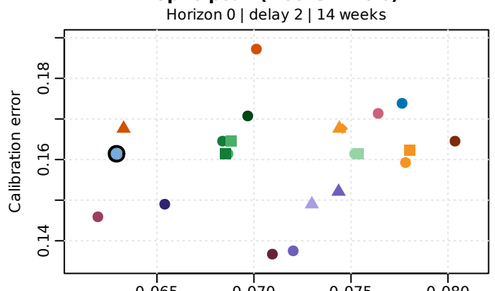


Feedback delay 2 week(s)

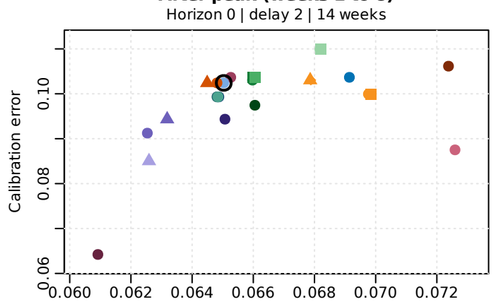
Jun 2025 to the end of season 2025/26
Horizon 0 | delay 2 | 68 weeks



Up to peak (weeks -7 to 0)
Horizon 0 | delay 2 | 14 weeks



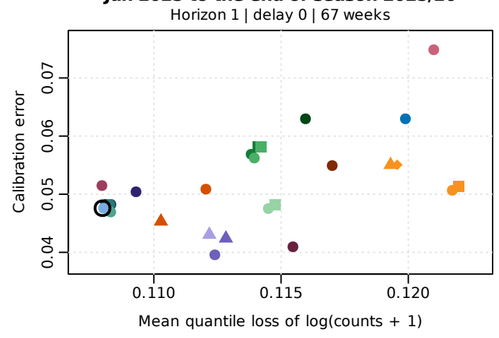
After peak (weeks 1 to 8)
Horizon 0 | delay 2 | 14 weeks



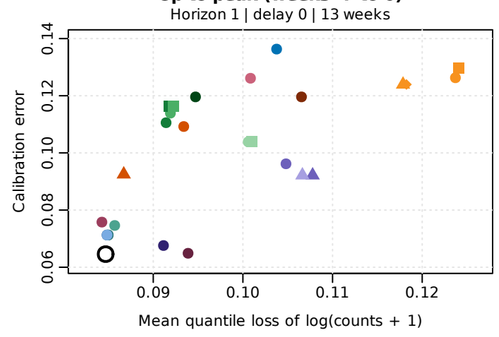
Horizon 1

Feedback delay 0 week(s)

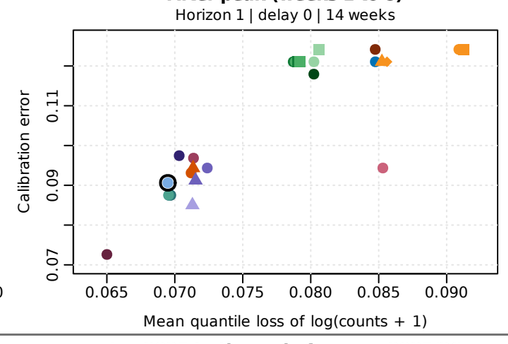
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 0 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 0 | 13 weeks

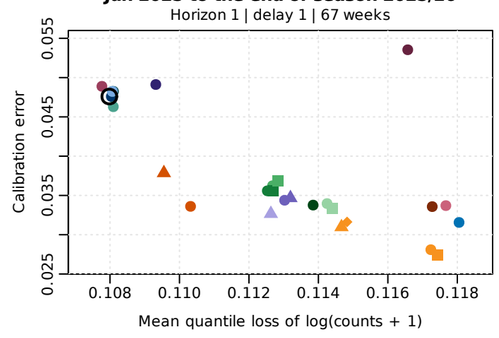


After peak (weeks 1 to 8)
Horizon 1 | delay 0 | 14 weeks

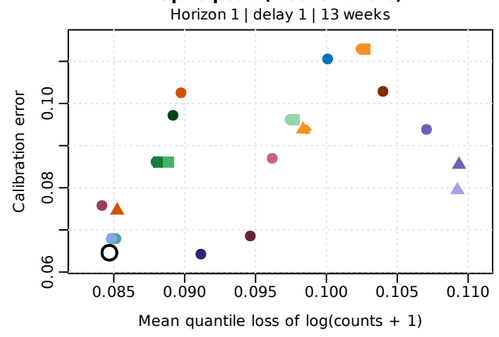


Feedback delay 1 week(s)

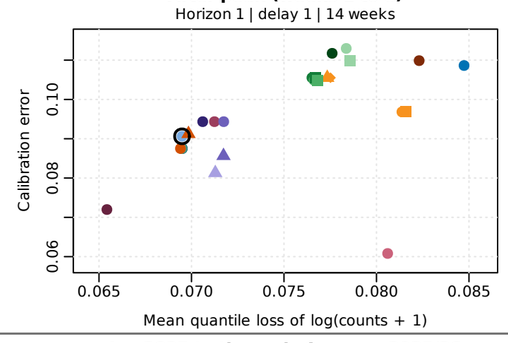
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 1 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 1 | 13 weeks

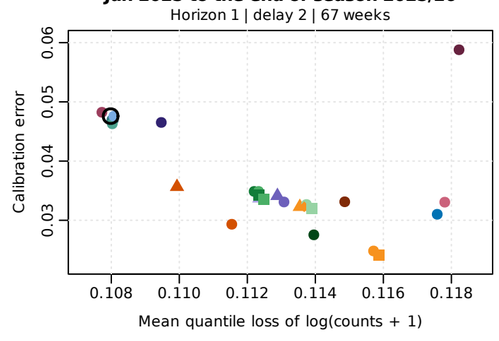


After peak (weeks 1 to 8)
Horizon 1 | delay 1 | 14 weeks

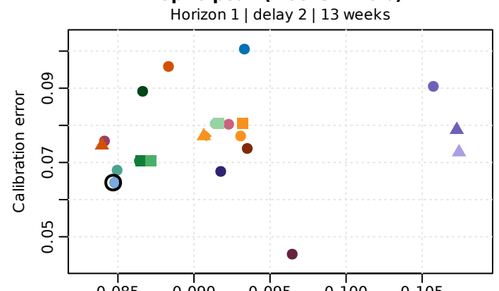


Feedback delay 2 week(s)

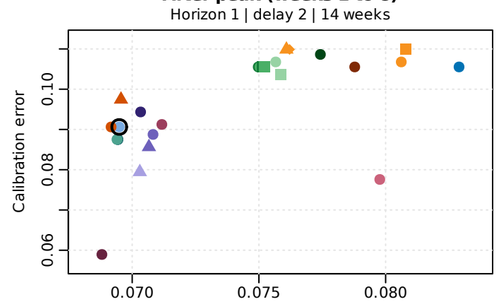
Jun 2025 to the end of season 2025/26
Horizon 1 | delay 2 | 67 weeks



Up to peak (weeks -7 to 0)
Horizon 1 | delay 2 | 13 weeks



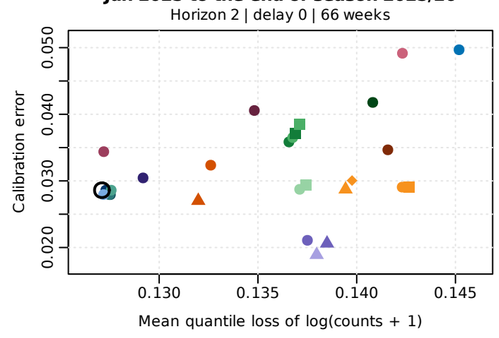
After peak (weeks 1 to 8)
Horizon 1 | delay 2 | 14 weeks



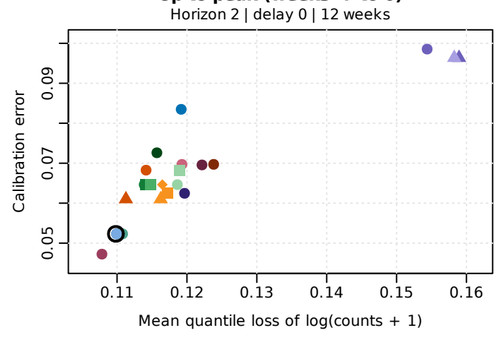
Horizon 2

Feedback delay 0 week(s)

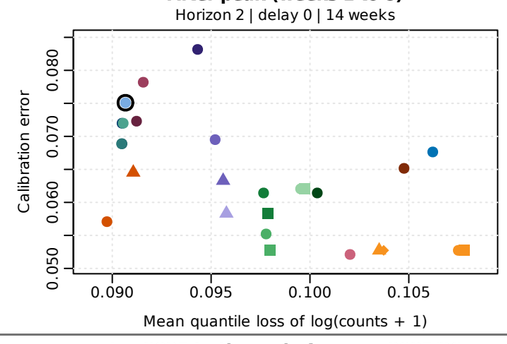
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 0 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 0 | 12 weeks

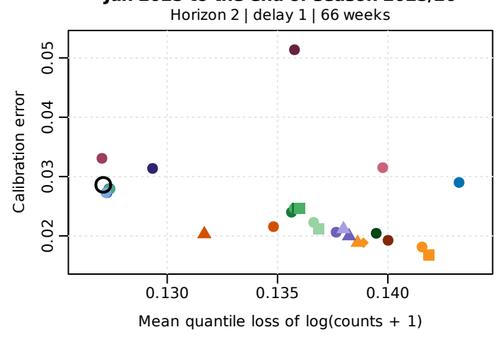


After peak (weeks 1 to 8)
Horizon 2 | delay 0 | 14 weeks

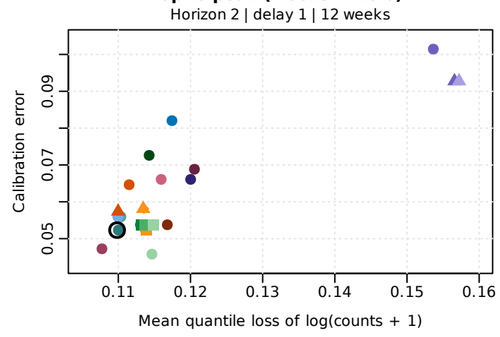


Feedback delay 1 week(s)

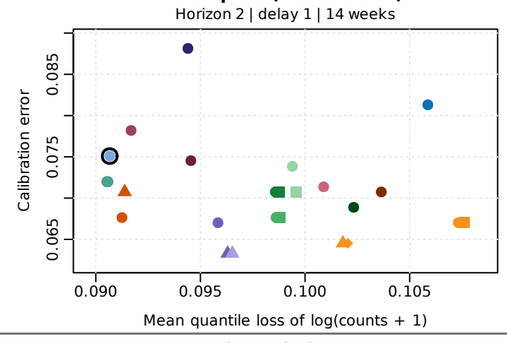
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 1 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 1 | 12 weeks

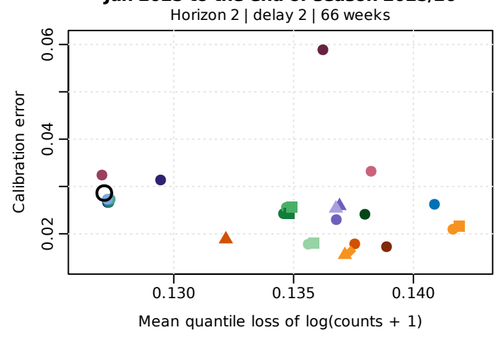


After peak (weeks 1 to 8)
Horizon 2 | delay 1 | 14 weeks

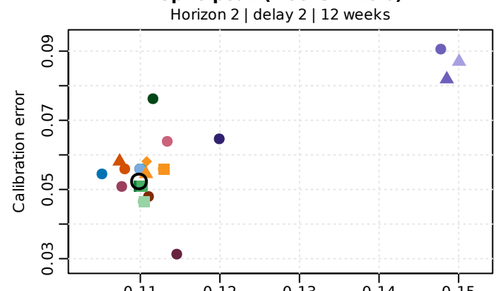


Feedback delay 2 week(s)

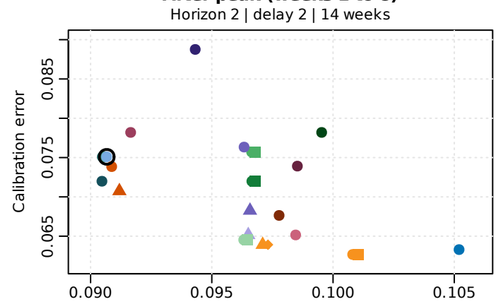
Jun 2025 to the end of season 2025/26
Horizon 2 | delay 2 | 66 weeks



Up to peak (weeks -7 to 0)
Horizon 2 | delay 2 | 12 weeks



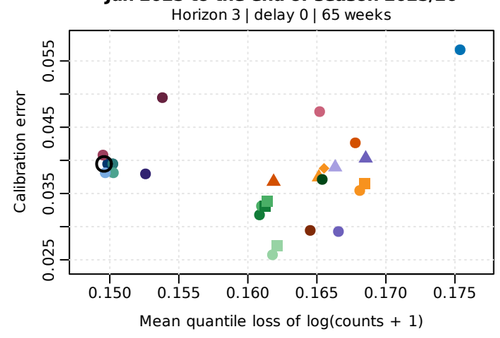
After peak (weeks 1 to 8)
Horizon 2 | delay 2 | 14 weeks



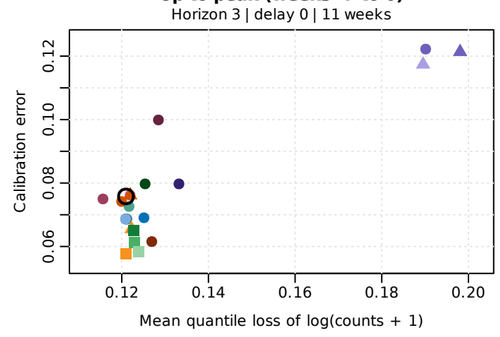
Horizon 3

Feedback delay 0 week(s)

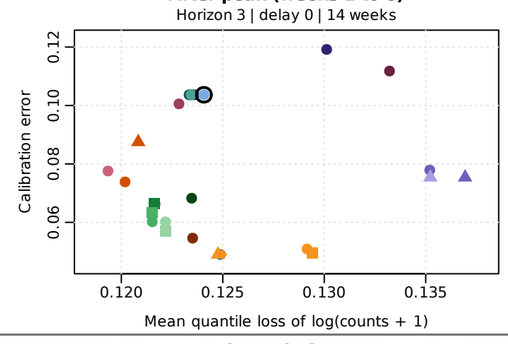
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 0 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 0 | 11 weeks

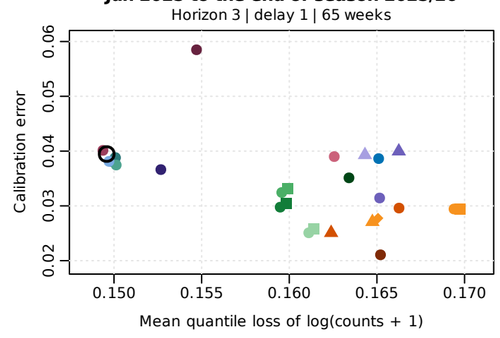


After peak (weeks 1 to 8)
Horizon 3 | delay 0 | 14 weeks

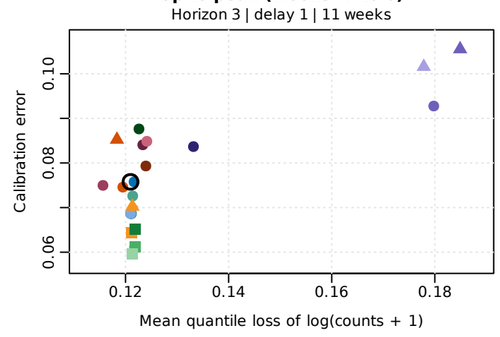


Feedback delay 1 week(s)

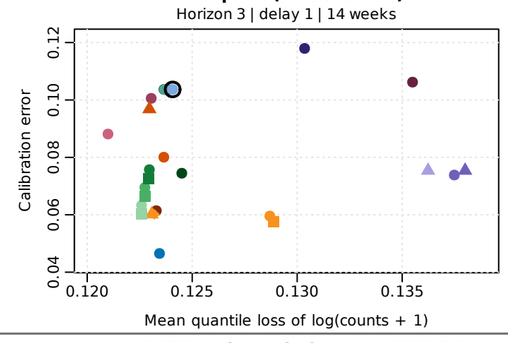
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 1 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 1 | 11 weeks

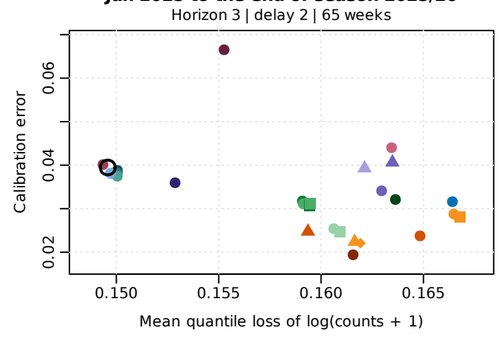


After peak (weeks 1 to 8)
Horizon 3 | delay 1 | 14 weeks

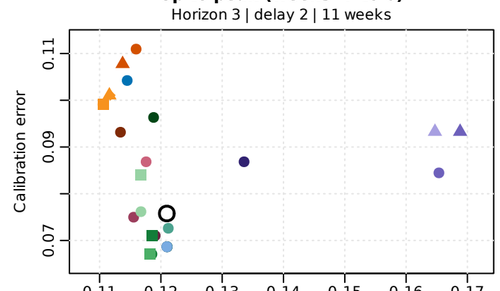


Feedback delay 2 week(s)

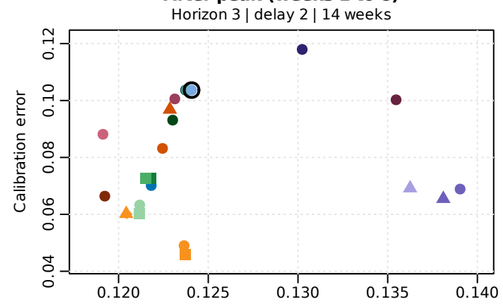
Jun 2025 to the end of season 2025/26
Horizon 3 | delay 2 | 65 weeks



Up to peak (weeks -7 to 0)
Horizon 3 | delay 2 | 11 weeks



After peak (weeks 1 to 8)
Horizon 3 | delay 2 | 14 weeks



Single trackers
- Original
- Fixed MultiQT (eta=0.001*S0)
- Log-slope rate (min m=0.002)
- Target oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=2)
- Rate grid: learned + sharing, common weights (H=2)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Diverse experts
- Diverse experts: equal (slow m=0.001)
- Diverse experts: learned + sharing (slow m=0.001, H=8)
- Diverse experts: learned + sharing, common weights (slow m=0.004)
- Diverse experts: soft switching (strength=1)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.003)
- Studentized median (base 80% width, normal start, eta=0.003)
- Median, constant sigma (RMS of 5, normal start, eta=0.1)

Studentized full base
- Studentized full base (base 80% width, eta=0.003)
- Studentized full base (level scales, eta=0.003)
- Studentized full base (level RMS of 5, eta=0.003)

Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the base's 80% interval width / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets start at the base; level scales = normal sd of the base's central interval at each level, or the RMS of each level's last 5 released residuals (weighted isotonic projection).